



GE Medical Systems

Technical Publications

Direction 2316677

Revision 1

Magnet and System Deinstallation for Upgrade to Signa® 1.5T *TwinSpeed*

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Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Immediately complete "**Damage Loss Claim Form**" after damage is found. Form is available via MS Exchange Mail at Outlook/Public Folder/All Public Folders/Medical Systems/!Global Initiatives/Information Management/Forms/Common Forms/Damage Loss Claim Form. Send completed form to e-mail address listed on form.

For more information about the Transportation Claim Procedure, access the GE Medical Systems Intranet and enter the following URL address:

<ftp://3.87.40.2/globepro/qualsys/Docs/190016ME.PDF>

Rev. 11/15/2000



GE Medical Systems

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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・ このサービスマニュアルには英語版しかありません。
- ・ GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・ このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・ この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
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Rev 0	.. Sep 20, 2001	Initial Release
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Rev 1	... Oct 2, 2002	Section 1 – Updated the material return process with GoldSeal as the main contact for disposition logistics. Added SRFD2 cabinet. Section 3 – Added RF/PEN cabinet to deinstall of 5.x and 8.x systems. Section 4 – Added cable removal for EXCITE Technology upgrade. Section 5 – Updated return forms by adding GEMS GoldSeal to top of page.
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Direction 2128126 ...	1*	3–1 to 3–4	0	4–1 to 4–8	1	5–7 to 5–8	0	5–17	1
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* This revision number/letter corresponds to the indicated document's revision control system.

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SECTION 1 – GETTING STARTED

1-1 SIGNA 1.5T TwinSpeed SYSTEM CONFIGURATION

This manual provides instructions to deinstall an existing magnet and system in preparation for upgrading to a new Signa® 1.5T TwinSpeed System. This deinstallation manual is applicable to field strengths of 1.5T.



FERROUS MATERIAL HAZARD! THE CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

1-2 DE-INSTALLED MATERIAL RETURN PROCESS

Getting the de-installed equipment sent back to the correct location is another essential step in the Magnet Upgrade process. The de-installed equipment may be sent back to a variety of places depending upon the type of upgrade being performed. Advance planning of equipment return will aid in the upgrade process. The three categories of equipment to be returned are as follows:

1. P & E hardware - i.e. InSite hardware, EDM Magnet Monitor Hardware.
2. System cables, cabinets, magnet enclosure, and miscellaneous hardware.
3. Magnet and associated magnet subsystem components.

For Magnet Monitor return (P&E Hardware) Refer to Direction 2108906, Magnet Monitoring Subsystem (Restricted), SECTION 7, DE-INSTALLATION PROCEDURES, for process and contacts for returning the equipment.

Note

All equipment being returned to GoldSeal will be sold as used equipment. Special care must be taken by the mechanical de-installers not to cut any cables or damage any of the hardware to be returned.

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

Note

The following equipment or material removed by the upgrade are **required to be returned** to GoldSeal:

- SGD or Gradient Cabinet
- System Cabinet (if removed)
- Indigo Computer with all associated boards/cables (includes reflex board, cable kit, and audio cables)
- All other replaced components

1-2 DE-INSTALLED MATERIAL RETURN PROCESS (continued)

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal.

The process for returning material is as follows:

1. Contact either of the following for disposition, with GoldSeal being the primary contact.
 - GoldSeal – Contact the MR Product Specialist at GoldSeal Logistics phone number 877–251–4468. If GoldSeal is accepting the returned equipment then all transportation arrangements and packaging will be processed through them. If GoldSeal is not accepting the return then they will advise you to contact GEMS Renewable Resources.
 - GEMS Renewable Resources – For coordination of transportation arrangements and packaging. Phone (414)–747–6997 or Dial Comm 8*579–6997; Fax (414)–747–6855; Field ASPEN 1–800–525–1516, Box #26041.
2. When calling, the following information is needed.
 - FDO (Future Delivery Order) number under which the upgrade was shipped.
 - Site name, address, and phone numbers.
 - Type of magnet being returned
 - Date of completed upgrade.
 - Name and Beeper number of Field Engineer that could be available when the equipment is picked up.
3. Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.
4. Process product locator information for all serialized components removed from the system according to policy.

1-3 DE-INSTALLED MATERIAL RETURN COORDINATION

The equipment will generally be returned on different trucks. The following lists may be useful in the collection and coordination of the equipment to be returned.

Prepare the return items for shipment as follows:

- a. Package other removed modules and boards in a cardboard box and secure inside System Cabinet (or any other cabinet where they fit).
- b. Use ramp to move cabinets on to shipping pallets. Three installers are required for this heavy lifting task. Fasten with shipping brackets that were removed previously from new cabinets.

Truck A - P & E Hardware

- ☐ EDM and mounting hardware
- ☐ InSite Modem and external Power Supply
- ☐ Magnet Monitor Modem and external Power Supply
- ☐ Modem Cables
- ☐ Magnet Monitor Sensor Kit - Water Flow/Temperature Sensor, Pressure Transducer, Reed Switch and Key Switch
- ☐ Magnet Monitor Cables (equipment room and magnet room)
- ☐ Magnet Monitor Interface Box, external AC/DC Power Supply and ribbon cables

Truck B - System Hardware**Note**

The list will vary depending on the type of electronics upgrade performed.

- ☐ System Cabinet
- ☐ RF Cabinet
- ☐ Gradient Cabinet
- ☐ Penetration Cabinet
- ☐ Penetration Panel
- ☐ PDU Cabinet
- ☐ Lytron - Heat Exchanger (air to water)
- ☐ Heat Exchanger Fluids (empty container and ethylene glycol label provided with upgrade hardware)
- ☐ Operator Console
- ☐ System Computer
- ☐ Magnet Enclosure Frame & mounting hardware
- ☐ Magnet Enclosure - All fiberglass parts and trim skirts
- ☐ Front & Rear Endbells
- ☐ Bridge & Carriage Cover
- ☐ Rear Pedestal & Electronic Components mounted within
- ☐ Rear Pedestal Covers
- ☐ Dock Assembly
- ☐ PAC Module
- ☐ SRI Module
- ☐ Patient Comfort Module

1-3 DE-INSTALLED MATERIAL RETURN COORDINATION (continued)

- ☐ RF/Gradient Body Coil
- ☐ System Cables removed from the Equipment Room
- ☐ System Cables removed from the Magnet Room
- ☐ System Cables removed from Operator Room
- ☐ Phantoms
- ☐ Non-conforming surface coils and extremity coils.
- ☐ Patient Table (optional - only if customer ordered new one and does not want the old one)

Truck C - Magnet & Subsystem Components**Note**

All Oxford and S-I magnets will be returned per instructions supplied by GEMS Renewable Resources for disposal.

- ☐ Magnet with cryogenics
- ☐ Cryogen Meters, instrumentation cables and power cable(s).
- ☐ ERU/MRU, instrumentation cable and power cable
- ☐ Magnet vent pipe and fittings
- ☐ Magnet Ramp Supply, Ramp Cables and Ramp Lead Extensions or Current Probes (Used on Oxford & S-I Magnets)
- ☐ Magnet Shim (Resistive) Power Supply and Shim Cables (Used on Oxford and optional for S-I, S-II, S-III and S-IV Magnets)
- ☐ Magnet Shim (Superconducting) Power Supply and Shim Cables (Used on S-I Magnets)
- ☐ Magnet Shim (Superconducting) Power Supply and Shim Cables (Used on S-I Magnets)
- ☐ Shield Cooler Compressor, Compressor Power Cable, Coldhead Power Cable, Supply & Return Gas Lines, and Remote Control Box (Balzers only).
- ☐ Magnet spares kit consisting of burst disc, baffles, vacuum release tool, o-rings, miscellaneous fittings etc.
- ☐ Magnet service ladder
- ☐ Magnet service platform
- ☐ Resistive Shim Filter Box on Penetration Panel
- ☐ Cable support bracket (S1 Magnet)
- ☐ Field mapping fixture

1-4 RETURN ITEMS CHECKLIST – BY ROOM

The following items are designated to be returned after the upgrade. All materials should be able to be packaged for return on the two provided cover dollies and pallets. Refer to Section 1-2, EQUIPMENT RETURN PROCESS for truck assignments.

Magnet Room

- ☐ 1. Magnet
- ☐ 2. Magnet Enclosure Front and Rear Cover
- ☐ 3. Magnet Enclosure Side Covers (4)
- ☐ 4. Magnet Enclosure Frame
- ☐ 5. Front End Bell
- ☐ 6. Rear End Bell
- ☐ 7. Left and Right Side Slide Support assemblies
- ☐ 8. Service Platform
- ☐ 9. Rear Pedestal Frame and covers
- ☐ 10. Gradient/RF Coil in shipping container
- ☐ 11. Bore Lamp Asm with Fiber Optic bundle
- ☐ 12. Fiberglass Bridge
- ☐ 13. Obsolete Cables
- ☐ 14. Coolant removed from Heat Exchanger and Body Coil (later systems only)
- ☐ 15. EDM Magnet Monitor Sensor Cables
- ☐ 16. ERU/MRU Module

Depending on the type of system being upgraded, some of the items listed below may not be returned. Refer to the appropriate Upgrade Flowchart in Section 3 for details.

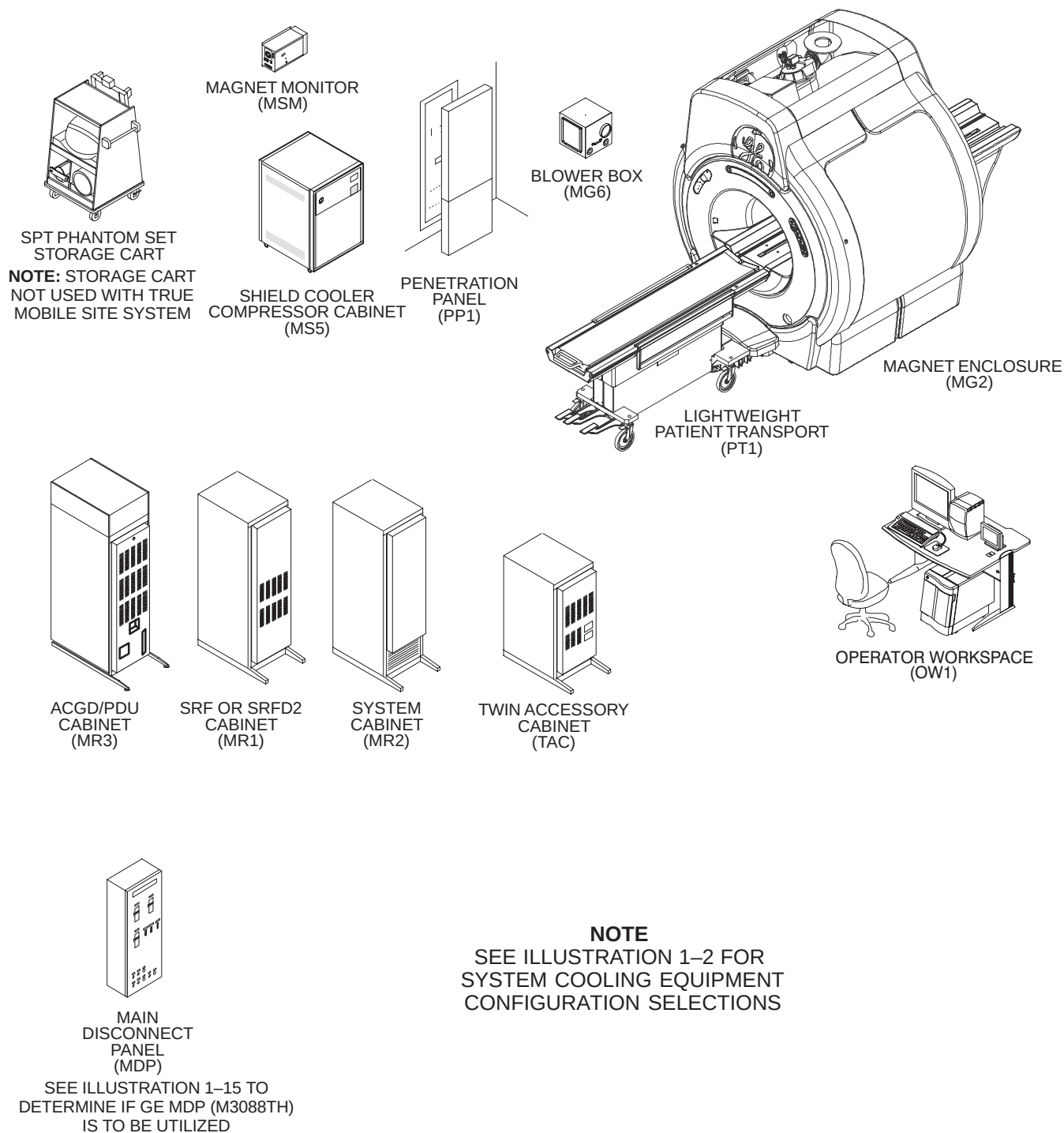
Equipment Room

- ☐ 1. Obsolete Cables
- ☐ 2. Penetration Panel
- ☐ 3. System Cabinet
- ☐ 4. Gradient (GRAM) Cabinet or SGD Cabinet
- ☐ 5. RF Cabinet or RF/PEN Cabinet
- ☐ 6. Penetration Cabinet
- ☐ 7. Full-Size PDU, Compact PDU, or RF/PDU
- ☐ 8. Computer Cabinet
- ☐ 9. Heat Exchanger (later systems only)
- ☐ 10. Water Chiller (later systems only)
- ☐ 11. Balzer/Leybold Shield Cooler Compressor, Gase Lines, and Cold Head Power Cables
- ☐ 12. EDM Magnet Monitor Computer, Water Flow Switch, and Cables

Operator Room

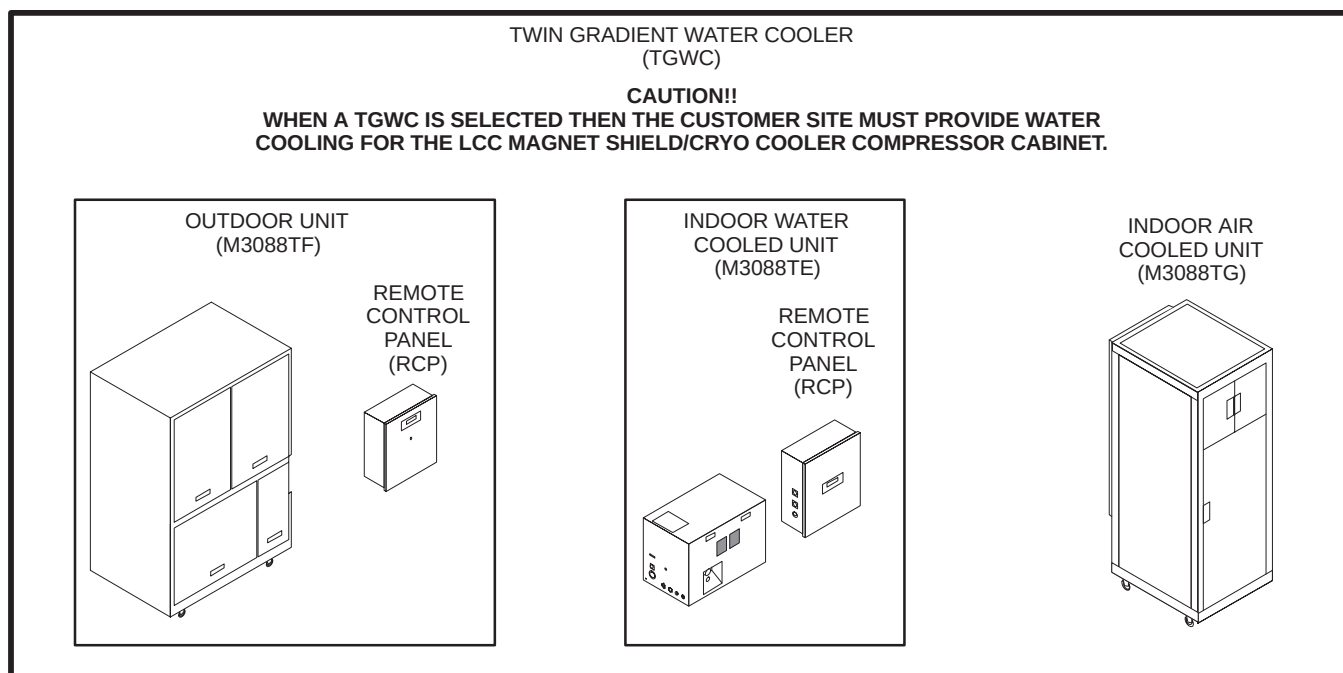
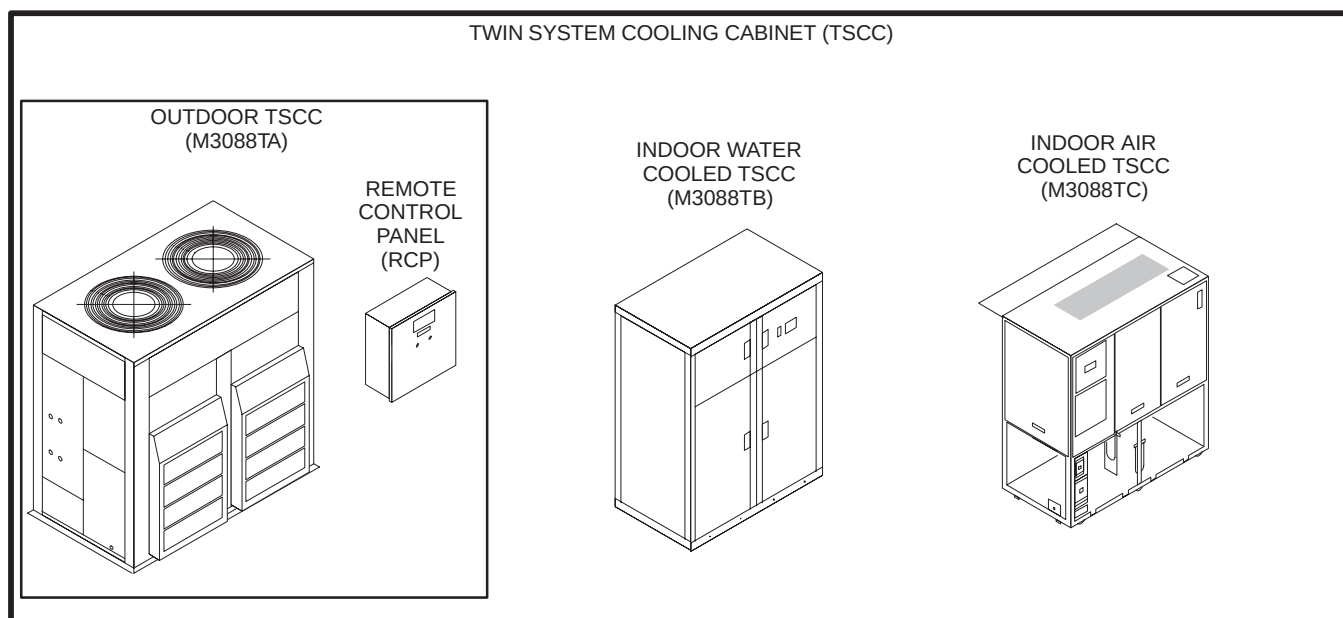
- ☐ 1. Console
- ☐ 2. Obsolete Cables
- ☐ 3. Balzer Remote Box

1-5 SYSTEM CONFIGURATION AFTER UPGRADE



SIGNA TwinSpeed SYSTEM AFTER UPGRADE

ILLUSTRATION 1-1

1-5 SYSTEM CONFIGURATION AFTER UPGRADE (continued)

SYSTEM COOLING EQUIPMENT SELECTIONS
ILLUSTRATION 1-2

SECTION 2 – POWER OFF PROCEDURE

2-1 SYSTEM POWER SHUTDOWN



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Verify that System performance baseline measurements have been obtained. and center frequency should be recorded before system is shut down.
2. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
3. Press the FULL OFF button on the PDU front control panel.



ELECTRIC SHOCK HAZARD!! HIGH VOLTAGES ARE PRESENT ON INTERCONNECTING CABLES USED WITH SPECIAL FIELD INSTALLED ION-PUMPS USED WITH OXFORD MAGNETS. THE ION PUMP POWER SUPPLY IS USUALLY CONNECTED TO A POWER SOURCE THAT IS NOT CONTROLLED BY THE PDU AND HIGH VOLTAGE WILL BE PRESENT EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK, DISCONNECT POWER TO THE ION PUMP POWER SUPPLY, LOCK OUT, AND TAG BEFORE PERFORMING PROCEDURES IN THE FOLLOWING SECTIONS. ALSO CHECK FOR OTHER INTERCONNECTING EQUIPMENT WHICH WAS FIELD INSTALLED AND DISCONNECT POWER, LOCK OUT, AND TAG.

4. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
5. Locate and shut off main disconnect supplying power to the PDU.
6. Locate and shut off main disconnect supplying power to the Shield Cooler Compressor.
7. If installed, locate and shut off main disconnect supplying power to the Magnet Monitor.
8. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
9. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring

SECTION 3 – SYSTEM UPGRADE DEINSTALLATION PROCESS

3-1 DEINSTALLATION FLOWCHARTS

This section contains information on:

<u>TITLE</u>	<u>PAGE</u>
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FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT, CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS
REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE
STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.
KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

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SIGNA 1.5T TwinSpeed UPGRADE FROM:

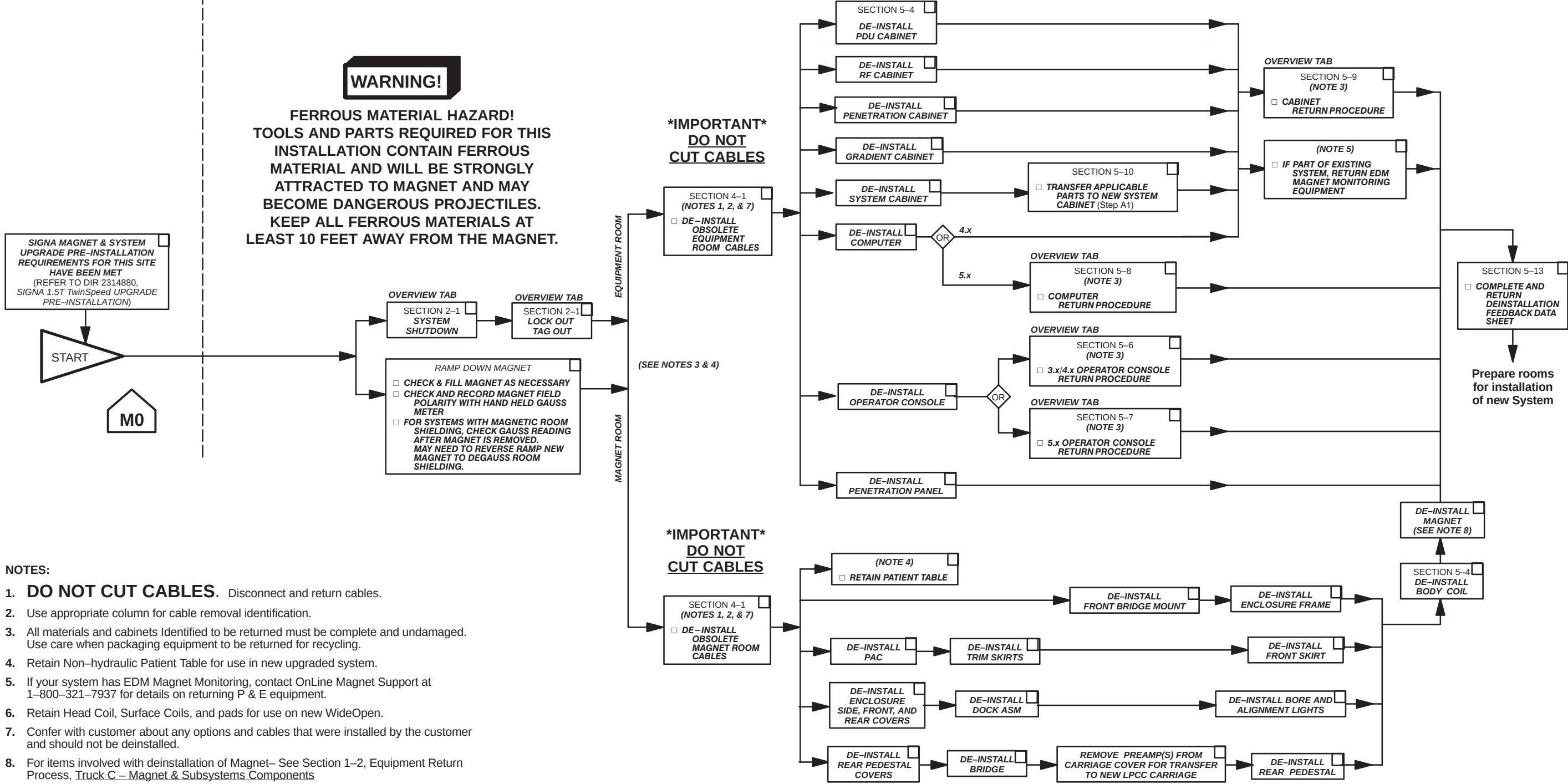
- SIGNA ADVANTAGE 4.x
- SIGNA ADVANTAGE 5.x 1.5T

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS

PRE-INSTALLATION

DE-INSTALLATION



SIGNA 4.x, and 5.x to SIGNA 1.5T TwinSpeed DEINSTALLATION FLOW CHART

ILLUSTRATION 3-1

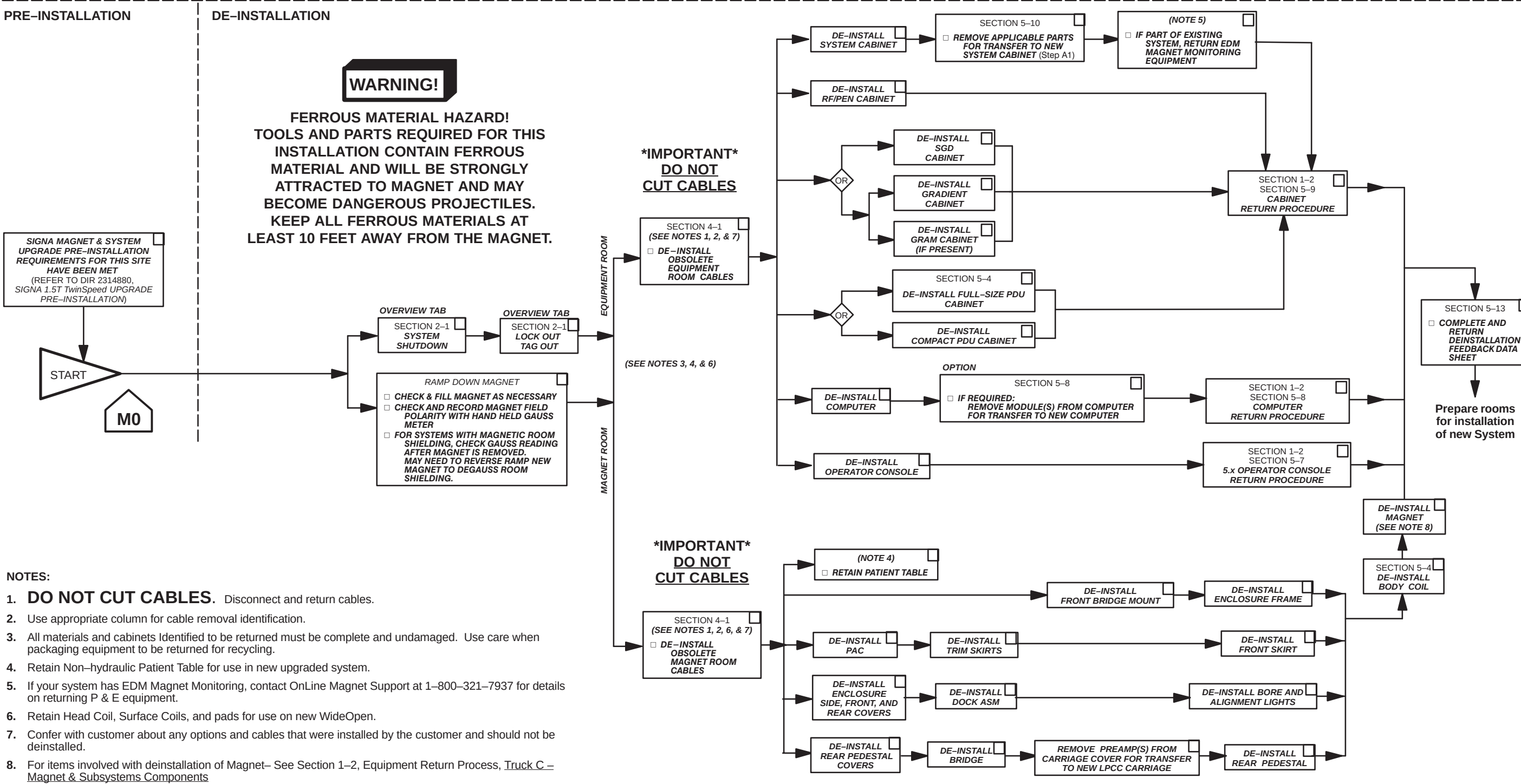
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SIGNA 1.5T TwinSpeed UPGRADE FROM:

- **SIGNA HORIZON 5.x**
 - with Full-Size or Compact PDU
 - with Gradient/GRAM or SGD

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS



SIGNA HORIZON 5.x to SIGNA 1.5T TwinSpeed DEINSTALLATION FLOW CHART

ILLUSTRATION 3-2

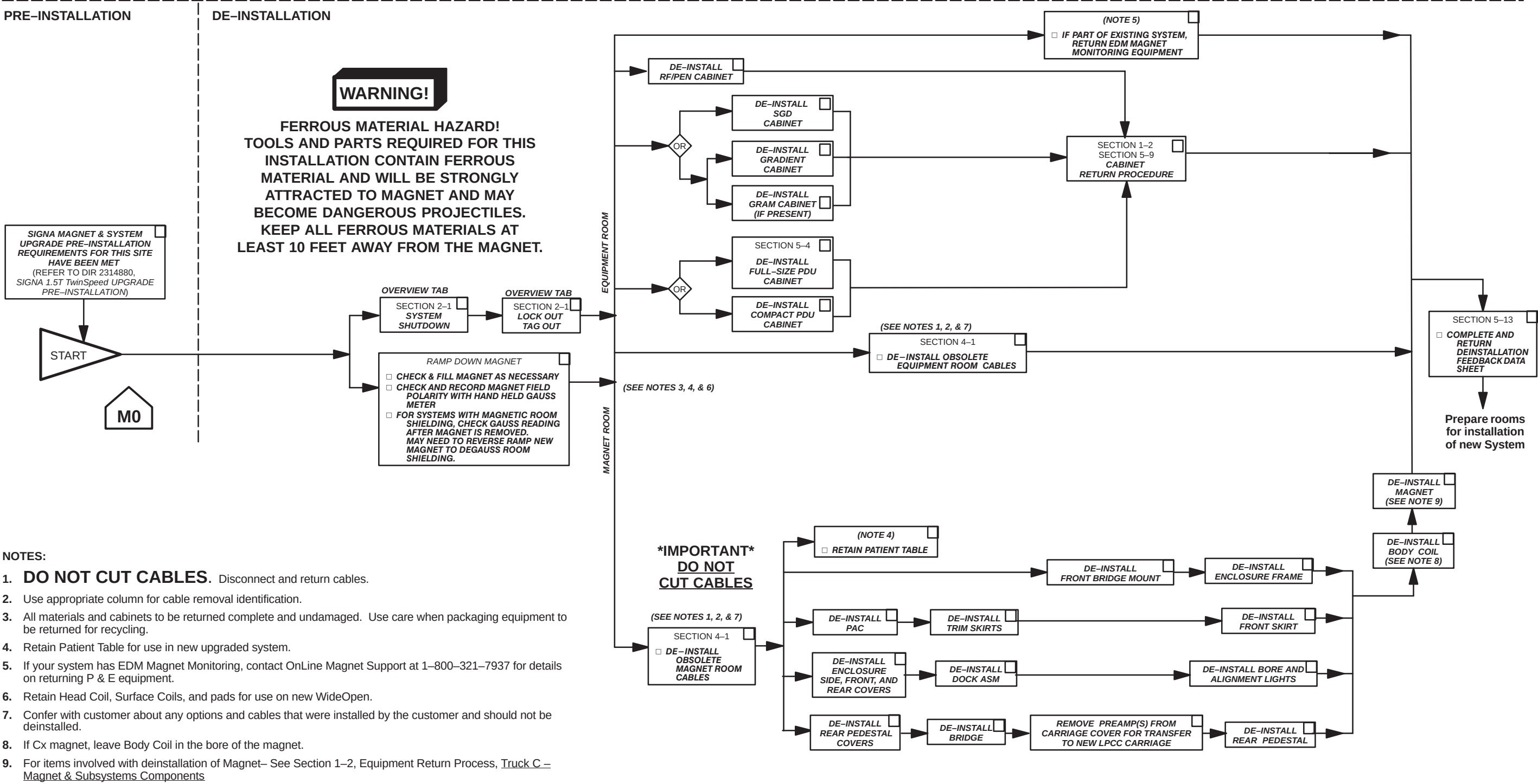
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SIGNA 1.5T TwinSpeed UPGRADE FROM:

- **SIGNA HORIZON 8.x**
 - with Full-Size or Compact PDU
 - with Gradient/GRAM or SGD

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS



NOTES:

1. **DO NOT CUT CABLES.** Disconnect and return cables.
2. Use appropriate column for cable removal identification.
3. All materials and cabinets to be returned complete and undamaged. Use care when packaging equipment to be returned for recycling.
4. Retain Patient Table for use in new upgraded system.
5. If your system has EDM Magnet Monitoring, contact OnLine Magnet Support at 1-800-321-7937 for details on returning P & E equipment.
6. Retain Head Coil, Surface Coils, and pads for use on new WideOpen.
7. Confer with customer about any options and cables that were installed by the customer and should not be deinstalled.
8. If Cx magnet, leave Body Coil in the bore of the magnet.
9. For items involved with deinstallation of Magnet– See Section 1-2, Equipment Return Process, Truck C – Magnet & Subsystems Components

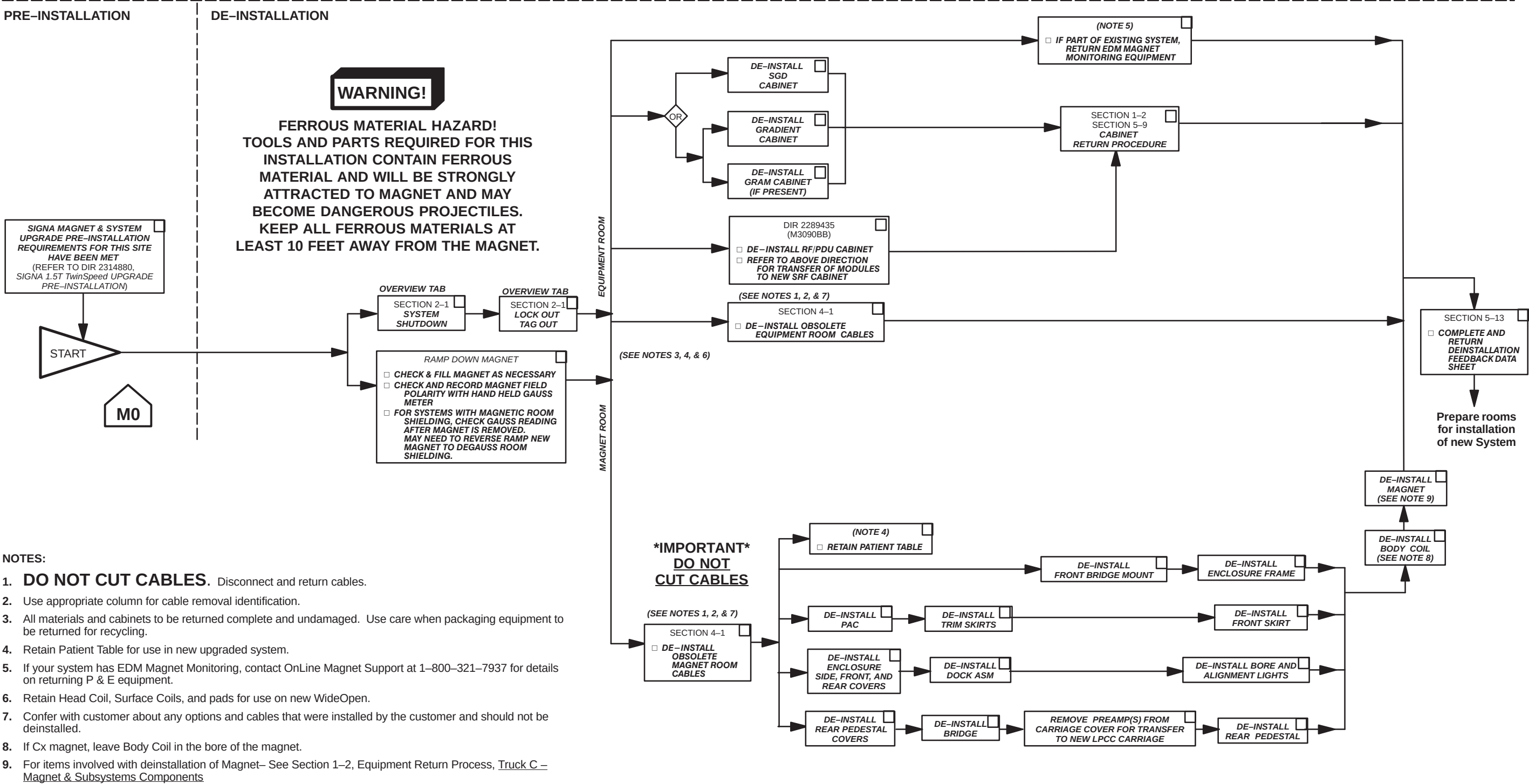
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SIGNA 1.5T TwinSpeed UPGRADE FROM:

- **SIGNA HORIZON 8.x**
 - with RF/PDU
 - with Gradient/GRAM or SGD

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS



SIGNA HORIZON 8.x to SIGNA 1.5T TwinSpeed DEINSTALLATION FLOW CHART
ILLUSTRATION 3-4

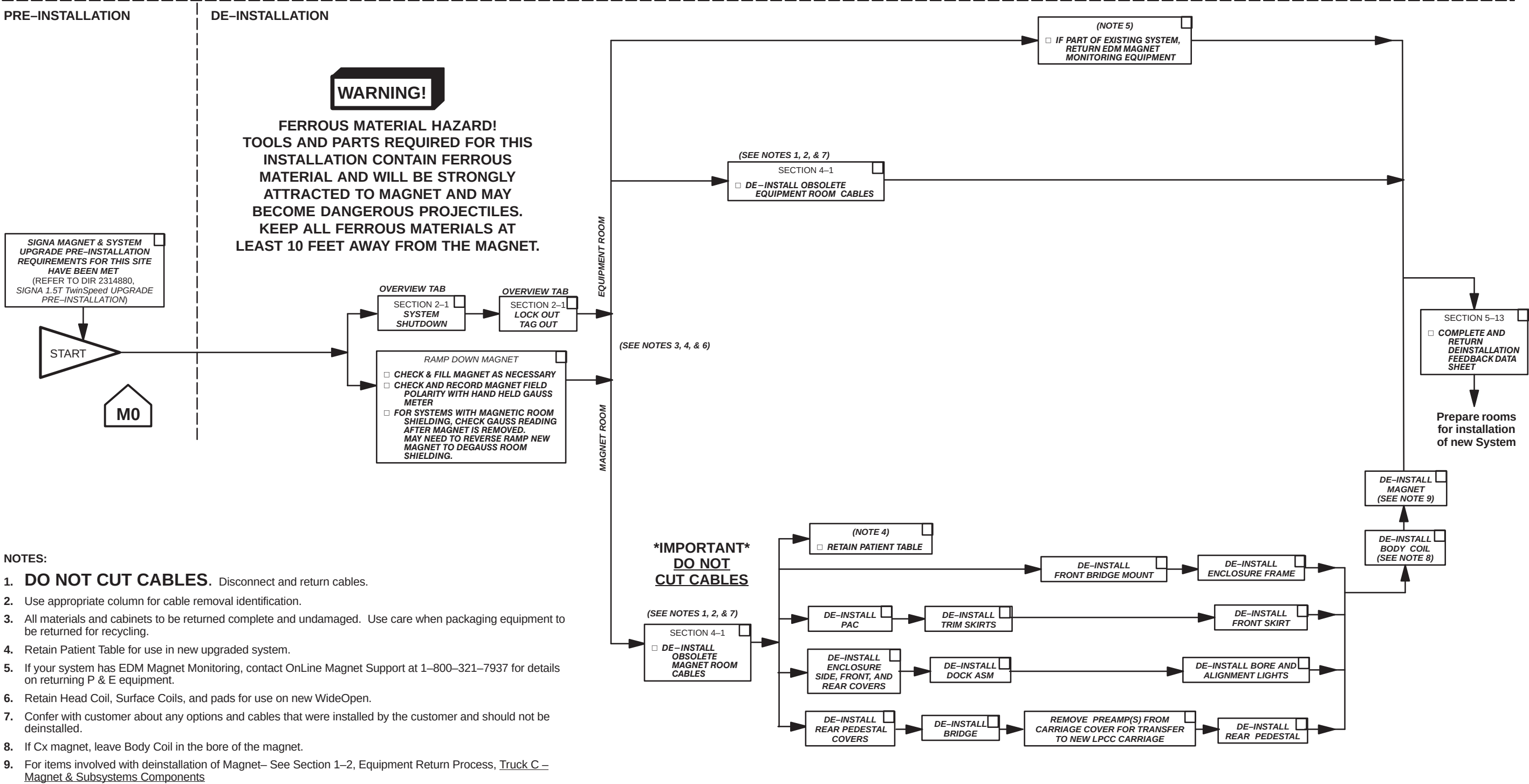
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SIGNA 1.5T TwinSpeed UPGRADE FROM:

- **SIGNA HORIZON 8.x**
– with ACGD/PDU

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS



SIGNA HORIZON 8.x to SIGNA 1.5T TwinSpeed DEINSTALLATION FLOW CHART
ILLUSTRATION 3-5

SECTION 4 – SYSTEM CABLES

This section provides information on removing system cables for upgrading a magnet and system to Signa 1.5T *TwinSpeed*.

4–1 CABLE REMOVAL FOR UPGRADING TO 1.5T *TwinSpeed* WITH EXCITE TECHNOLOGY (MGD)

If site is also being upgraded to a system with EXCITE Technology, then refer to Direction 2333499, *Signa System Upgrade for MGD*, (M3090EB) for instructions on cable removal as well as System cabinet and other modules and equipment that require upgrading. The EXCITE upgrade is performed along with the applicable upgrade in Section 4–1.

4–2 CABLE REMOVAL FOR UPGRADING TO 1.5T *TwinSpeed*

Table 4–1 lists all cables to be removed from various early system configurations, including Advantage 4.x, Advantage 5.4, Horizon 5.x, and Horizon 8.x with non–LCC magnets, for upgrading to a Signa 1.5T *TwinSpeed* system. Signa 4.x and pre–5.4 Signa Advantage systems that are being upgraded to Signa 1.5T *TwinSpeed* require almost all cables to be removed. **DO NOT CUT CABLES.** Disconnect and return cables.

The letters on the right side of the header on table 4–1 are defined by the following cable removal explanations:

- A** – Existing system is Signa Horizon 8.x with ACGD/PDU & SRF Cabinet upgrading to *TwinSpeed*
- B** – Existing system is Signa Horizon 8.x with RF/PDU & SGD Cabinet upgrading to *TwinSpeed*
- C** – Existing system is Signa Horizon 8.x with Full–Size/Compact PDU & SGD upgrading to *TwinSpeed*
- D** – Existing system is Signa Horizon 8.x with Full–Size/Compact PDU & Gradient/GRAM upgrading to *TwinSpeed*
- E** – Existing system is Signa Horizon 5.x with SGD Cabinet upgrading to *TwinSpeed*
- F** – Existing system is Signa Horizon 5.x with Gradient/GRAM cabinets upgrading to *TwinSpeed*
- G** – Existing system is 5.4 Signa Advantage upgrading to *TwinSpeed*
- H** – Existing system is 4.5 or later but pre 5.4 Signa Advantage upgrading to *TwinSpeed*
- I** – Existing system is 4.x but pre 4.5 Signa Advantage upgrading to *TwinSpeed*

The upgrades in this manual will have many new cables delivered to the site. In some cases the new cables are the same as the old cables being removed. These are indicated in the table by the letters **E** and **M** and are defined by the following explanations:

- E**= Equipment Room Cable. A new cable with the same Run number will be shipped to site for upgrade to *TwinSpeed*. Site FE must determine whether to keep existing cable or remove it and install new cable.
- M** = Magnet Room Cable. A new cable with the same Run number will be shipped to site for upgrading to *TwinSpeed*. Site FE must determine whether to keep existing cable or remove it and install new cable.
- X** = Cable to be removed and returned for recycling. **DO NOT CUT CABLES.**

4-2 CABLE REMOVAL FOR UPGRADING TO 1.5T TwinSpeed

TABLE 4-1
CABLES TO BE REMOVED AND RETURNED WHEN UPGRADING TO 1.5T TwinSpeed

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I
002	PD1-A2-TS1-32/NEUT/GND	MR2-A12-SERVICE								X	X
003	PD1-A2-TS1-1,2,3/NEUT/GND	MR3-A10-AC IN							X	X	X
007	PD1-FEEDER 15	PP2-A6-AC IN								X	X
008	PD1-FEEDER 16	PP2-A6-SERVICE								X	X
009	PD1-A2-TS1-10,11,12/NEUT/GND	MR1-J1			X	X	X	X	X	X	X
012	PD1-A2-TS1-21/NEUT/GND	OC1-A8-J3 (AC IN)								X	X
013	PD1-A2-TS1-33/NEUT/GND	OC1-A8-J6 (SERVICE)								X	X
014	PD1-A2-TS1-34/NEUT/GND	CC1-A20-SERVICE								X	X
015	PD1-A2-TS1-13,14,15/NEUT/GND	CC1-A20-AC IN								X	X
016	PD1-A2-TS1-16,17,18/NEUT/GND	CC2-A20-AC IN								X	X
017	PD1-A2-TS1-34/NEUT/GND	CC2-A20-SERVICE								X	X
021	PD1-A2-GND	CC1-A20-GND								X	X
022	PD1-A2-GND	CC2-A20-GND								X	X
023	PD1-GND BUS	OC1-GND					X	X	X	X	X
027	PD1-A2-TS1-32/NEUT/GND	PP2-A15-J1-SERVICE								X	X
028	PD1-A2-TS1-31/NEUT/GND	PP2-A15-J2-AC IN								X	X
030	PD1-A2-TS1-16,17,18/NEUT/GND	MR2-A12-J1 AC IN		X	X	X	X	X	X		
035	PD1-A2-TS1-22,23,24/NEUT/GND	CC2-A20-AC IN								X	X
036	PD1-FEEDER 5-TS1-4,5,6 (Spect.)	MR6 (Spectroscopy Only)							X	X	X
037	PD1-GND	MR2-A12-GND	E	X	X	X	X	X	X		
038	PD1-GND BUS	MR1-A7 GND	E		X	X	X	X	X		
040	MS1	RF COM GND	X	X	X	X	X	X	X		
041	PD1-A2-TS1-1,2,3,NUETGND	MR3-A11-J9		X	X	X	X	X			
042	PD1-A2-TS1-4,5,6,GND	MR3-A12		X	X	X	X	X			
043	MR7-A4-J7	PD1-A2				X		X			
044	MR1-A7-GND	PP1-GND	E	E	E	E	X	X			
045	PP1-GND	FAN MODULE GND	M	M	M	M	X	X			
046	HE1	PD1-TS1-20,NUE,GND			X	X	X	X			
047	PD1-A2-TS1-25,31 OR 21,15	OW1-A2-A5-J1		X	X	X					
048	PD1-GND	OW1-A2-A5-GND	E	E	E	E					
050	OC1-AC-IN	PD1-TS1-13,14,15,NUE,GND					X	X	X		
064	MR2-A12-J1-AC OUT	CC1					X	X	X		
065	WC1	PD1-A2-TS1-26,27,GND		X							
200	MR2-A11-J19	MR3-A9-J1								X	X
201	MR2-A11-J20	MR3-A9-J2								X	X
202	MR2-A11-J21	MR3-A9-J3								X	X
203	MR2-A11-J22	MR3-A9-J4								X	X
223	OC1-A8-J13	MR2-A1-J36								X	X
224	CC1-A21-J19	MR1-A7-J8								X	X
226	MR1-A7-J5	MR2-A11-J14								X	X
227	PD1-A1-A2-J2	MR2-A11-J16							X		

4-2 CABLE REMOVAL FOR UPGRADING TO 1.5T TwinSpeed

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I
228	CC1-A21-J10	MR2-A11-J17								X	X
229	MR1-A7-J3	MR2-A11-J1	E	E	E	E	X	X	X		
231	PP1-J8	MR2-A11-J2	E	E	E	E	X	X	X		
235	PP1-J44	MR1-A7-J13	E	E	E	E	X	X	X		
236	PP1-J4	MR1-A7-J12	E	E	E	E	X	X	X		
238	CC1-A21-J14	OC1-A8-J8								X	X
239	CC1-A21-J15	OC1-A8-J10								X	X
244	CC1-A21-J13	BTI-HOST								X	
248	CC2-A21-J1	OC1-A9-J1								X	X
249	CC2-A21-J22	OC1-A8-J12								X	X
256	PP1-J12	OC1-A8-J11								X	X
257	PP1-J44	MG3-A3-A1-J1	M	M	M	M	X	X	X	X	X
258	PP1-J4	MG3-A17-J1	M	M	M	M	X	X	X	X	X
261	PP1-J6	MG2-A11-A1-J1							X	X	X
262	PP1-J8	MG3-A3-A5-OUT	M	M	M	M	X	X	X	X	X
274	CC1-A21-J20	MR1-A7-J7								X	X
280	PP1-J15	PP2-A1-J2								X	X
281	PP1-J11	PP2-A1-J3								X	X
282	PP1-J12	MG3-A2-J6	M	M	M	M	X	X	X	X	X
287	CC2-A21-J15	CC1-A21-J2								X	X
296	PP1-J18	EO2	E	E	E	E	X	X	X		
297	PP1-J18	EO1									
300	PP1-J15	MG2-A15	X	X	X	X	X	X	X	X	X
301	CC2-A21-J20	CC1-A21-J28								X	X
302	CC2-A21-J21	CC1-A21-J29								X	X
318	MG3-A7	PP1-J11	M	M	M	M	X	X	X	X	X
322	MR2-A11-J11	PP2-A6-J18								X	X
325	CC1-A21-J27	CC2-A21-J19								X	X
327	CC1-A21-J25	CC2-A21-J17								X	X
346	MR2-A11-J41	CM1-FRONT								X	X
347	MR2-A11-J42	CM1-BACK								X	X
354	CM1	ST1								X	X
370	PP1-A9-J46	MR1-A7-J19									X
384	PP1-J20	PP2-A15-J17							X	X	X
385	PP1-J20	MG2-A30	X	X	X	X	X	X	X	X	X
386	PP1-J47	PP2-A15-J21							X	X	X
387	PP1-J47	MG2-A29	M	M	M	M	X	X	X	X	X
388	PP1-J11	PP2-A15-J22							X	X	X
389	PP1-J15	PP2-A15-J20							X	X	X
404	MG3-A17-J1	MG2-A16-A5-J6	X	X	X	X	X	X	X	X	X
405	MR3-A9-TS1	MG3-TS1-1,2							X	X	X
406	MR3-A9-TS1	MG3-TS1-3,4							X	X	X
407	MR3-A9-TS1	MG3-TS1-5,6							X	X	X
408	PD1-A1-A3-J3	PP1-J16							X	X	X
409	PP1-J16	MG2-A12-A1-J7							X	X	X

4-2 CABLE REMOVAL FOR UPGRADING TO 1.5T TwinSpeed

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I
410	PP1-J14	PP2-A15-J23									X
411	PP1-J14	MG3-A13-J1									X
412	PP1-J51	MG3-A32-J1							X	X	X
413	PP1-J51	PP2-A15-J28(A16-J9)							X	X	X
414	MG3-A35	MG3-A2-J9	X	X	X	X	X	X	X	X	X
415	PP1-A8-J45	MG3-A33-J7									X
416	MR1-A7-J21	PP1-A8-J45									X
417	PP2-A15-J19	MR2-A11-J32								X	X
418	MR2-A11-J39	PP2-A15-J3									X
419	MR2-A11-J26	PP1-A10-J1								X	X
420	MG3-A33-J1	PP1-A10-J2								X	X
421	MG2-A33-J3	MG3-A31-J1	X	X	X	X	X	X	X	X	X
422	MG2-A33-J2	MG3-A1-J4	X	X	X	X	X	X	X	X	X
424	PP1-A10-J5/J6	MG2-A33-J11/J12								X	X
425	MR2-A11-J12	PP1-J6									X
426	MR1-A7-J9	MR3-A9-J5								X	X
427	MR2-A11-J17	CC1-A21-J13								X	X
428	MR2-A11-J44	CC1-A21-J35								X	X
429	BT1	CC1-A21-J25								X	X
430	MR2-A11-J24	CC1-A21-J26								X	X
431	MR2-A11-J25	CC1-A21-J27								X	X
432	PD1-A1-A3-J2	MR2-A11-J45							X		
433	OC1-A9-J1	CC1-A21-J10								X	X
434	CC1-A8-J12	CC1-A21-J29								X	X
435	MR2-A11-J13	PP1-A10-J3								X	X
436	MG2-A12-A1-J1	MG3-A33-J3							X	X	X
437	MG2-A12-A1-J2	MG3-A33-J4							X	X	X
439	MG3-A3-J4	MG3-A33-J1							X	X	X
440	MG3-A3-J3	MG3-A34-J2							X	X	X
441	MG3-A33-J7	MG3-A34-J8							X	X	X
446	PP1-A9-J46	MG2-A12-A13-J5									X
447	PP1-A9-J46	MG2-A12-A13-J3									X
448	MR2-A11-J44	CC1-A21-J1								X	
449	MR2-A11-J24	CC1-A21-J34								X	
450	MR2-A11-J25	CC1-A21-J35								X	
453	PP1-J50	MG2-A11-A1-J4								X	X
454	MR2-A11-J34	PP1-J50								X	X
470	MR1-A7-J17	PP1-A11-J71							X	X	
474	PP1-A11-J72	MG3-A12-A1-J3							X	X	
475	PP1-A11-J73	MG3-A12-A1-J4							X	X	
476	PP1-A11-J74	MG3-A33-J7							X	X	
477	PP1-A11-J75	MG3-A12-A1-J5							X	X	
478	PP1-A11-J76	MG3-A12-A1-J6							X	X	
485	MG3-A17-J2	MG2-A16-A3-OUT	X	X	X	X	X	X	X	X	X
486	PP1-A14-J85	MR1-A7-J22	E	E	E	E	X	X	X		

4-2 CABLE REMOVAL FOR UPGRADING TO 1.5T TwinSpeed

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I
487	PP1-A15-J78	MR1-A7-J20	E	E	E	E	X	X	X		
488	PP1-J80	MR2-A11-J8	E	E	E	E	X	X	X		
489	PP1-J81	MR2-A11-J9	E	E	E	E	X	X	X		
490	PP1-J82	MR2-A11-J10	E	E	E	E	X	X	X		
491	PP1-J80	MG3-A36-J7	M	M	M	M	X	X	X	X	X
492	PP1-J81	MG3-A36-J8	M	M	M	M	X	X	X	X	X
493	PP1-J82	MG3-A36-J9	M	M	M	M	X	X	X	X	X
494	PP1-A14-J87	MG3-A36-J6	M	M	M	M	X	X	X	X	X
495	PP1-A15-J78	MG3-A36-J10	M	M	M	M	X	X	X	X	X
499	PP1-A14-J84	MR2-A11-J5	E	E	E	E	X	X	X		
536	WC1 (Coolant Line)	MG2-A12-A1	X	X	X	X	X	X			
537	WC1 (Coolant Line)	MG2-A12-A1	X	X	X	X	X	X			
601	MS1 (RED)	MS6 (RED)	X	X	X	X	X	X	X	X	X
602	MS1 (BLK)	MS6 (BLK)	X	X	X	X	X	X	X	X	X
603	MS1-A13-A1	MS7	X	X	X	X	X	X	X	X	X
604	MS1-A13-A1	MS6/MS7	X	X	X	X	X	X	X	X	X
605	PP1-J10 (Magnet Room)	MS1-A3-A1 (Magnet Room)	X	X	X	X	X	X	X	X	X
605	PP1-J10 (Equipment Room)	MR1-J11 (Equipment Room)	X	X	X	X	X	X	X	X	X
606	MS4	MS1-A3-A1	X	X	X	X	X	X	X	X	X
611	PP1-A5	MS3							X		
612	PP1-A5	MS1-A8							X		
621	MS5-A1 (Gas Line)	MS1-A2 (Gas Line)	X	X	X	X	X	X	X	X	X
622	MS5-A1 (Gas Line)	MS1-A2 (Gas Line)	X	X	X	X	X	X	X	X	X
623	MS5-A1	PP1-F1,F2,F3,F4	X	X	X	X	X	X	X	X	X
624	MS1-A2	PP1-F1,F2,F3,F4	X	X	X	X	X	X	X	X	X
626	MS5-A1	BALZER REMOTE		X	X	X	X	X	X	X	X
700	PP1-A14-J86	MG3-A17-J2		M	M	M	X	X	X		
702	MR2-A11-J18	MR1-A7-J5	E	E	E	E	X	X			
703	MR2-A11-J23	PD1-A14-A1-J2	E	X	X	X	X	X			
704	PP1-J12	OC1-A10-J9					X	X	X		
705	MR2-A11-J25	PP2-A15-J19							X		
706	MR2-A11-J14	PD1-A14-A1-J3	E	X	X	X	X	X	X		
707	MR2-A11-J26-ESTOP	OC1-J8					X	X	X		
708	MR2-A11-J16	MR1-A7-J27	E	E	E	E	X	X	X		
709	MR1-A7-J26	MR3-A9-J1	E	X	X	X	X	X	X		
710	MR2-A11-J17	MR3-A9-J2	E	X	X	X	X	X	X		
711	PP1-A17-A1-J91	MG2-A33	M	M	M	M	X	X	X		
711	PP1-A17-A1-J91	MR2-A11-J13	E	E	E	E	X	X	X		
712	PP1-A17-A1-J91	MG2-A11-A1	M	M	M	M	X	X	X		
712	PP1-A17-A1-J91	MG2-A11-J13	E	E	E	E	X	X	X		
713	PP1-J14	PP2-A15-J23							X		
714	PP1-A17-J1	PP2-A15-J27							X		
715	PP1-J14	MG2-A33-J1	X	X	X	X	X	X	X	X	X
716	PP1-J89	MG2-A11-A1-J2	X	X	X	X	X	X	X	X	X
718	MR2-A11-J29	OC1-A10-J4					X	X	X		

4-2 CABLE REMOVAL FOR UPGRADING TO 1.5T TwinSpeed

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I
719	CC1-A1-A6-J1	MR2-A15-A3-J1					X	X	X		
726	PP1-A16-J77	MR1-A7-J18	E	E	E	E	X	X	X		
727	PP1-A16-J77	MG3-A36-J5	M	M	M	M	X	X			
728	MG3-A36-J11	MG2-A16-OUT-J1	X	X	X	X	X	X	X	X	X
729	MG3-A36-J12	MG2-A16-OUT-J2	X	X	X	X	X	X	X	X	X
730	MG3-A36-J13	MG2-A16-OUT-J3	X	X	X	X	X	X	X	X	X
731	MG3-A36-J14	MG2-A16-OUT-J4	X	X	X	X	X	X	X	X	X
732	MG3-A36-J15	MG2-A16-OUT-J5	X	X	X	X	X	X	X	X	X
733	MG3-A36-J16	MG2-A16-OUT-J6	X	X	X	X	X	X	X	X	X
734	MG3-A36-J17	MG2-A16-OUT-J7	X	X	X	X	X	X	X	X	X
735	MG3-A36-J18	MG2-A16-OUT-J8	X	X	X	X	X	X	X	X	X
736	MG3-A36-J19	MG2-A16-OUT-J9	X	X	X	X	X	X	X	X	X
739	MG3-A3-A1-J4	MG2-A12-A1-J1							X		
740	MG3-A3-A1-J3	MG2-A12-A1-J2							X		
743	MG3-A3-J3	MG2-A12-J2	X	X	X	X	X	X			
744	MG3-A3-J4	MG2-A12-J1	X	X	X	X	X	X			
745	PP1-J44	MR1-A7-J13					X	X			
748	PP1-J4	MR1-A7-J12					X	X			
749	MG2-A36	MG2-A33-J10	X	X	X	X	X	X	X		
753	MR7-A4-J1	MG3-A11-J5				X		X			
754	MR7-A4-J2	MG3-A11-J4				X		X			
755	MR7-A4-J3	MG3-A11-J3				X		X			
756	MR7-A4-J4	MG3-A11-J8				X		X			
757	MR7-A4-J5	MG3-A11-J7				X		X			
758	MR7-A4-J6	MG3-A11-J6				X		X			
759	MR3-A11-J12	MR7-A4-J8				X		X			
760	MR3-A11-J11	MR7-A4-J9				X		X			
761	MR3-A11-J10	MR7-A4-J10				X		X			
762	MG3-A32-TS1/2	PP1-A7-1/2	X	X	X	X	X	X			
763	MG3-A32-TS3/4	PP1-A7-3/4	X	X	X	X	X	X			
764	MG3-A32-TS5/6	PP1-A7-5/6	X	X	X	X	X	X			
767	MR1-A7-J37 (See Note 1)	PP1-J51						X			
768	MR1-A7-J31	PP1-J20	E	E	E	E	X	X			
769	MR1-A7-J34	PP1-J15	E	E	E	E	X	X			
770	MR1-A7-J41	PP1-J11	E	E	E	E	X	X			
771	MR1-A7-J36	PP1-J47	E	E	E	E	X	X			
772	MR1-A7-J40	PP1-J14	E	E	E	E	X	X			
773	MR1-A7-J35	PP1-A17-J1	E	E	E	E	X	X			
774	MR1-A7-J32	MR2-A11-J25	E	E	E	E	X	X			
775	MR1-A7-J42	PP1-A11-J92	E	E	E	E	X	X			
776	MR1-A7-J44	PP1-A11-J93	E	E	E	E	X	X			
777	MR1-A7-J43	PP1-A11-J94	E	E	E	E	X	X			
778	MG2-A12-A13-J3	PP1-A11-J72	M	M	M	M	X	X			
779	MG2-A12-A13-J4	PP1-A11-J73	M	M	M	M	X	X			
780	MG3-A3-J6	PP1-A11-J74	M	M	M	M	X	X			

4-2 CABLE REMOVAL FOR UPGRADING TO 1.5T TwinSpeed

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I
781	MG2-A12-A12-J5	PP1-A11-J75	M	M	M	M	X	X			
782	MG2-A12-A12-J6	PP1-A11-J76	M	M	M	M	X	X			
783	PP1-F5/F6	MR1-A7-J37	E	E	E	E	X	X			
784	PP1-F5/F6	MG6-J1	M	M	M	M	X	X			
785	PP1-F7/F8	MR1-A7-J47	X	X	X	X	X	X			
786	PP1-F7/F8	MG6-J2	X	X	X	X	X	X			
787	MR2-A11-J30	CC1-A1-A1-SERIAL C					X	X			
788	OW1-A1-A4-J7	PP1-J12	E	E	E	E					
789	OW1-A1-A4-J18	MR2-A11-J26	E	E	E	E					
790	OW1-A2-A6-BIT3 (See Note 2)	MR2-A11-J21									
791	OW1-A16-PORT1	MR2-A11-J30	E	E	E	E					
800	OC1-J1	CC1-A2-J1 (MUX)					X	X	X		
801	OC1-J10	CC1-A2-J2 (SECUR. KEY)					X	X	X		
802	OC1-J12	CC1-A1-J9 (PLASMA)					X	X	X		
803	OC1-J13	CC1-A1-A10 (VIDEO)					X	X	X		
804	OC1-J3	CC1-A1-A1 (BOOT)					X	X	X		
818	OW1-A16-PORT6,7,8	MR2-A11-J31	E	E	E	E					
823	MSM1-A1-J10 (See Note 3)	MR1-A11-J24		X							
824	PP1-J10 (See Note 3)	FJ1		X							
825	PP1-J48 (See Note 3)	MSM1-A1-J8		X							
826	MSM1-A1-J9 (See Note 3)	FJ3		X							
827	FJ3 (See Note 3)	MS5-A5-A1-J1/FJ4		X							
828	PP1-J10 (See Note 3)	MS1-A3-A1-P403		X							
829	PP1-J48 (See Note 3)	MS1-FJ2		X							
830	MS1-FJ2 (See Note 3)	MS1A2-J2/MS1A5A2-J1/TB1		X							
831	FJ1 (See Note 3)	MSM1-A1-J7		X							
832	MS1-A2-J1 (See Note 3)	MS1-SLEEVE/COLDHEAD		X							
833	FJ4 (See Note 3)	MS5-A1-A6-JR		X							
836	MR2-A11-J21 (See Note 2)	OW1-A15-BIT3	E	E	E	E					
966	MR1-A7-J1	MR3-PD1-7,8,9,GND	E								
967	MR1-CB1	MR3-PD1-10,11,12,GND	E								
968	MR2-A11-J1	MR3-PD1-4,5,6,GND	E								
969	OW1-A5-J1	MR3-PD1-18,19,GND,NEUT	E								
970	WC1	MR3-PD1-TS1-16,17,GND	X								
971	MR2-A11-J23 (See Note 4)	PD1-A14-A1-J2 (See Note 4)	E	X	X	X					
972	MR2-A11-J14 (See Note 4)	PD1-A14-A1-J3 (See Note 4)	E	X	X	X					

Notes:

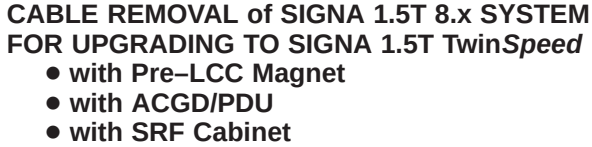
1. Run 767 is only found on a few early Horizon 5.x (5.5) systems.
2. Early versions of 8.x (LX) systems installed Run 790 for Bit3 connection to J21 on the system cabinet. Later versions installed Run 836. DO NOT REMOVE EITHER CABLE. Cable will be reused for this upgrade.
3. Magnet Monitor cables may not be present on all systems (option). If installed, remove cables and module.
4. Cables are double labeled: Run 971 is also Run 703, Run 972 is also Run 706.

4-3 CABLE REMOVAL MAPS

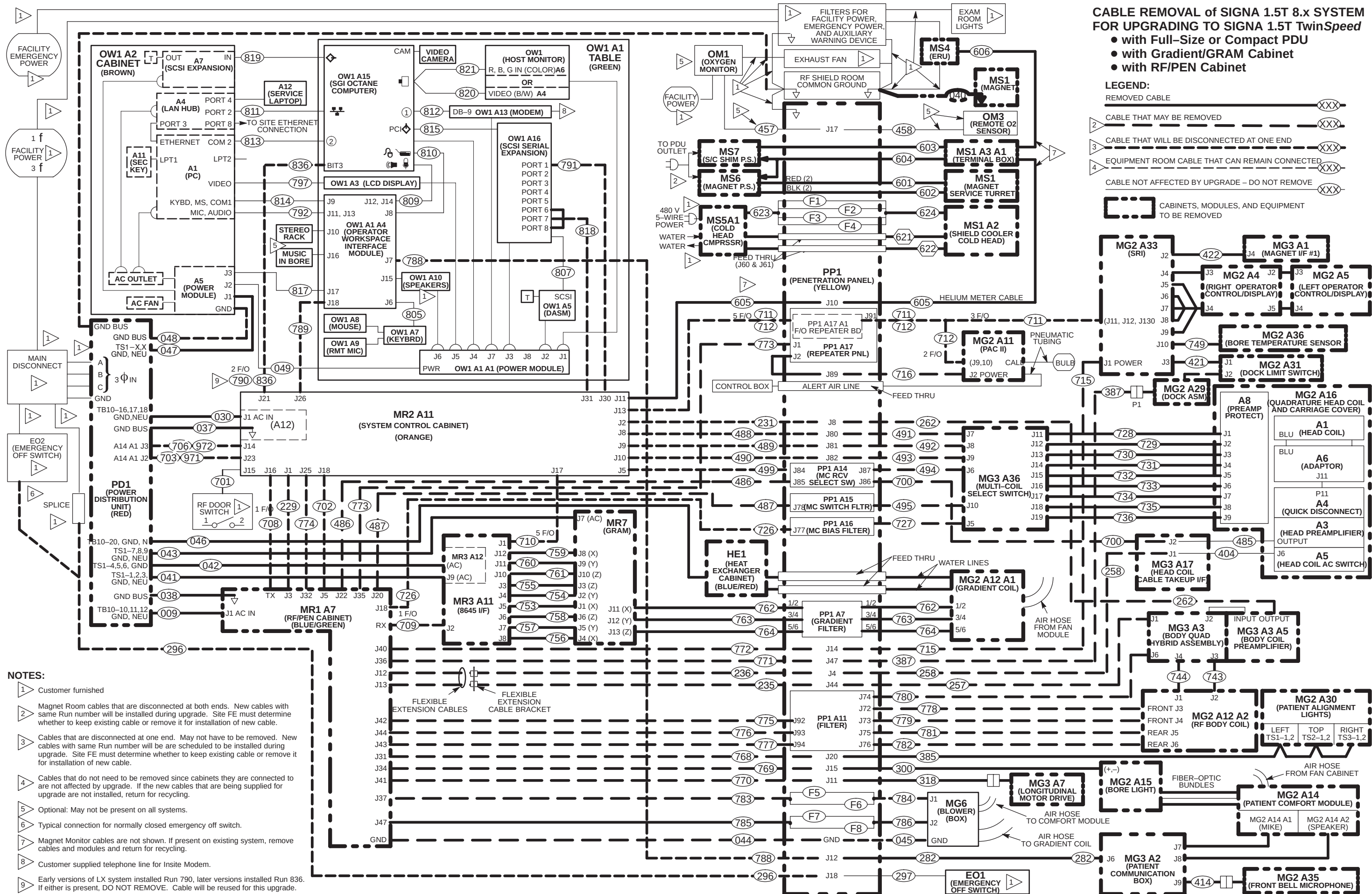
The following “Cable Removal” interconnect maps refer to cables removed when upgrading:

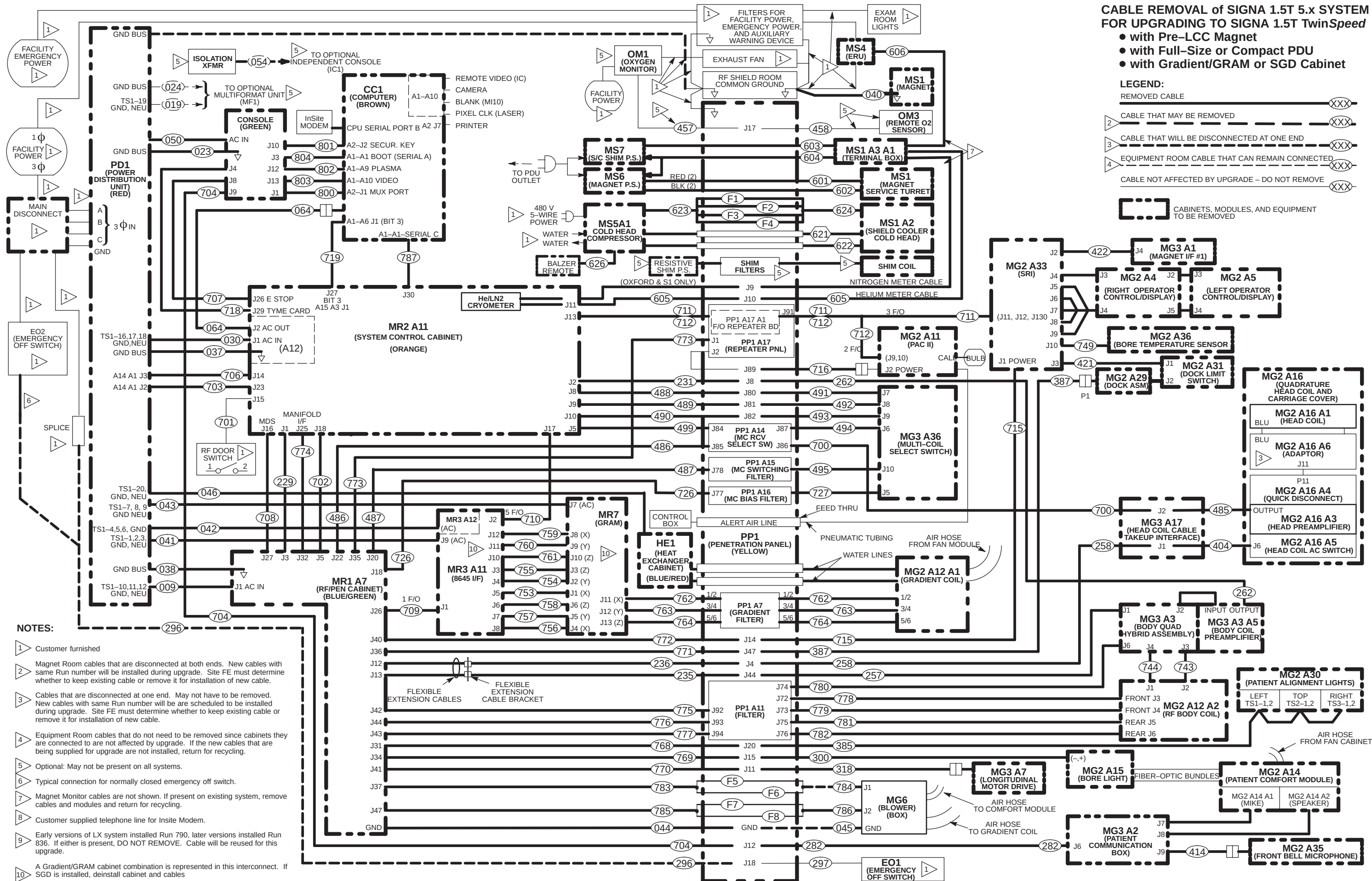
- Signa Horizon 8.x system with ACGD/PDU to TwinSpeed Page 4-9
- Signa Horizon 8.x system with RF/PDU and SGD to TwinSpeed Page 4-10
- Signa Horizon 8.x system with Full-Size/Compact PDU and SGD to TwinSpeed Page 4-11
- Signa Horizon 8.x system with Full-Size/Compact PDU and Gradient/Gram to TwinSpeed .. Page 4-12
- Signa Horizon 5.x system with Full-Size/Compact PDU and SGD or G/G to TwinSpeed Page 4-13
- Signa Advantage 5.4 system to TwinSpeed Page 4-14

Refer to Direction 2333499, *Signa System Upgrade for MGD*, (M3090EB) for the “Cable Removal” interconnect map if system is also being upgraded with EXCITE Technology.



- 3 Customer furnished
- 2 Magnet Room cables that are disconnected at both ends. New cables with same Run number will be installed during upgrade. Site FE must determine whether to keep existing cable or remove it for installation of new cable.
- 3 Cables that are disconnected at one end. May not have to be removed. New cables with same Run number will be scheduled to be installed during upgrade. Site FE must determine whether to keep existing cable or remove it for installation of new cable.
- 4 Equipment Room cables that do not need to be removed since cabinets they are connected to are not affected by upgrade. If the new cables that are being supplied for upgrade are not installed, return for recycling.
- 5 Optional: May not be present on all systems.
- 6 Typical connection for normally closed emergency off switch.
- 7 Magnet Monitor cables are not shown. If present on existing system, remove cables and modules and return for recycling.
- 8 Customer supplied telephone line for Insite Modem.
- 9 Early versions of LX system installed Run 790, later versions installed Run 836. If either is present, DO NOT REMOVE. Cable will be reused for this upgrade.





SECTION 5 – DEINSTALLATION

5-1 DEINSTALL MAGNET ENCLOSURE

The following are basic guidelines for all deinstallations:

- A. **DO NOT CUT CABLES.** Cables are to be disconnected and returned intact.
- B. Deinstalled cabinets, penetration panel, console, and any other modules or assemblies are to be returned intact.
- C. All fastening hardware needs to be returned for reuse. In all cases, after disassembly, loosely re-install all fastening hardware in mounting holes or threaded holes where they were originally located.
- D. Retain on site the Head Coil, all Surface Coils, and pads for reuse on new *TwinSpeed* system.

5-2 DEINSTALLATION FEEDBACK FORM

Located on the last page of this section is the Deinstallation Feedback Data Sheet. Fill out this form and send it to the address located on the bottom of the sheet. Any and all information provided will help improve the deinstallation process.

5-3 RETURN MATERIAL INFORMATION FORMS

Fill in the information on “Attention: GEMS GoldSeal” form for Genesis Console (page 5–9), Genesis Computer (page 5–13), and any deinstalled Cabinets (page 5–17) attach the appropriate sheet to each item.



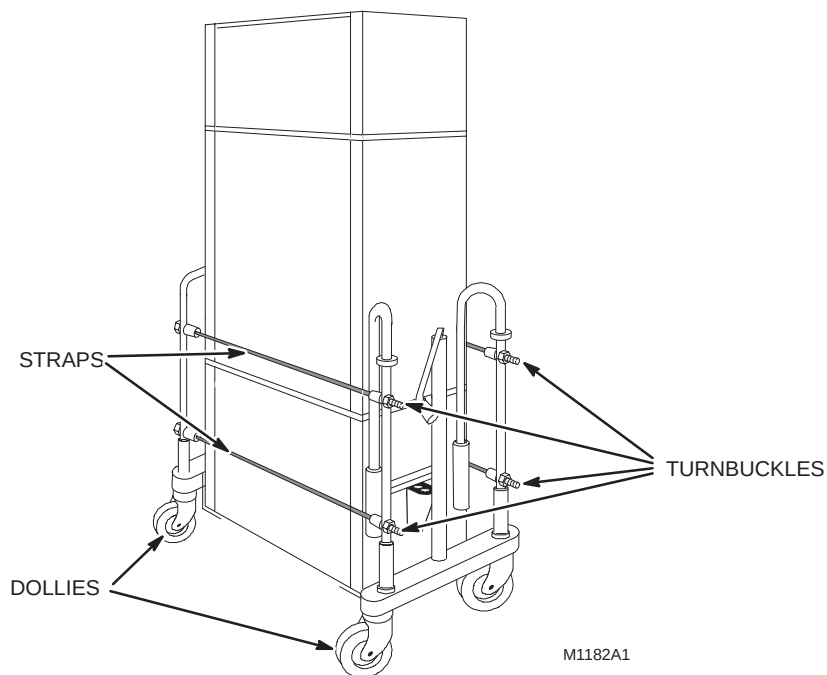
FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

5-4 REMOVING FULL-SIZE PDU

Sites that are upgrading to ACGD require that the full-size PDU or Compact PDU be de-installed. The Compact PDU has built-in casters for ease of movement. The full-size PDU does not have casters and will have to be moved with two shipping dollies placed on each side of the PDU and clamped in place using straps with turnbuckles as shown in illustration below. If dollies are not provided, the Full-size PDU will have to be removed by riggers or other locally supplied equipment moving personnel.



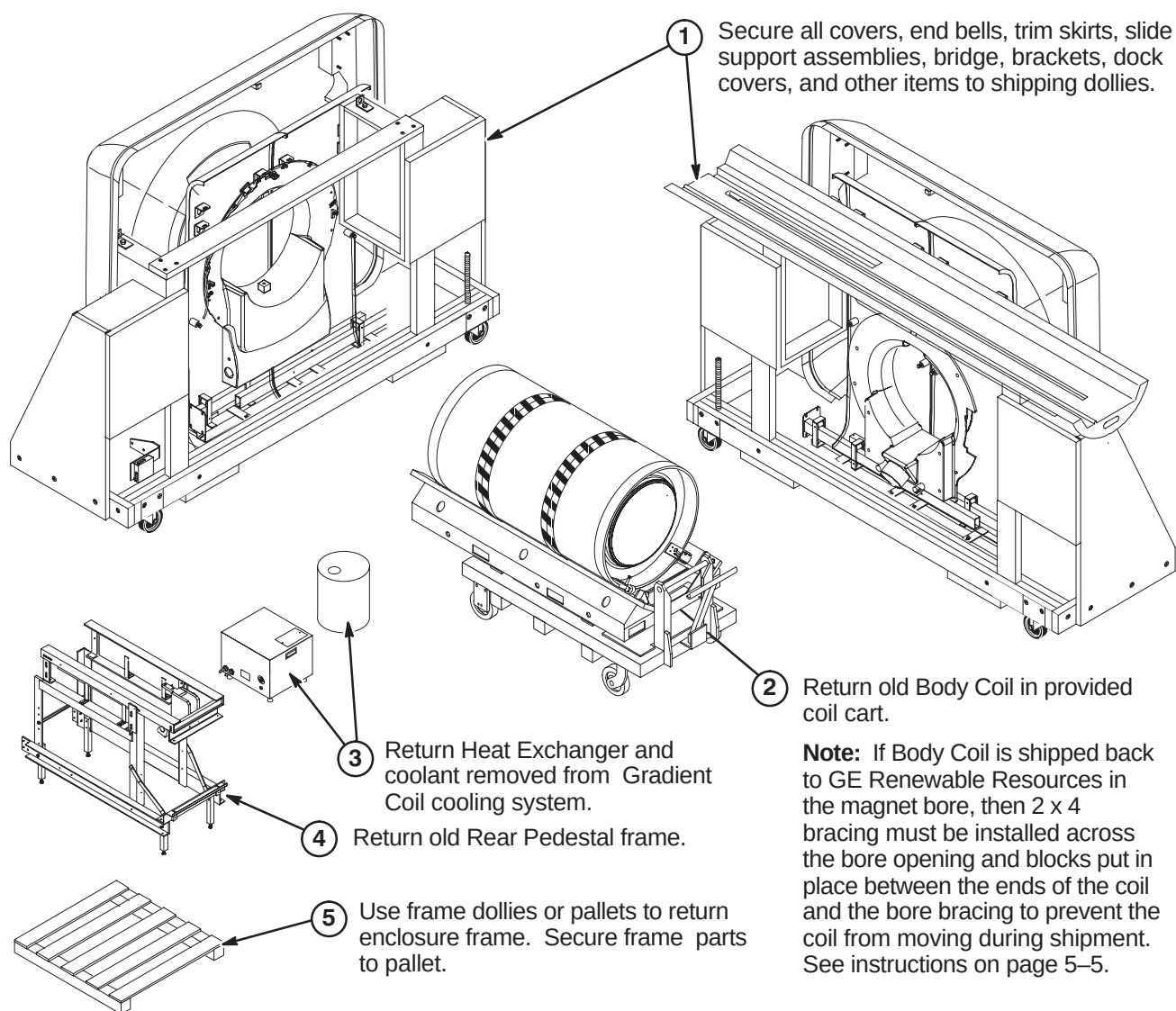
PDU weight is about 1000 pounds (455 kilograms). Plywood or steel plates may be used to distribute weight and protect floor surface from overload damage.



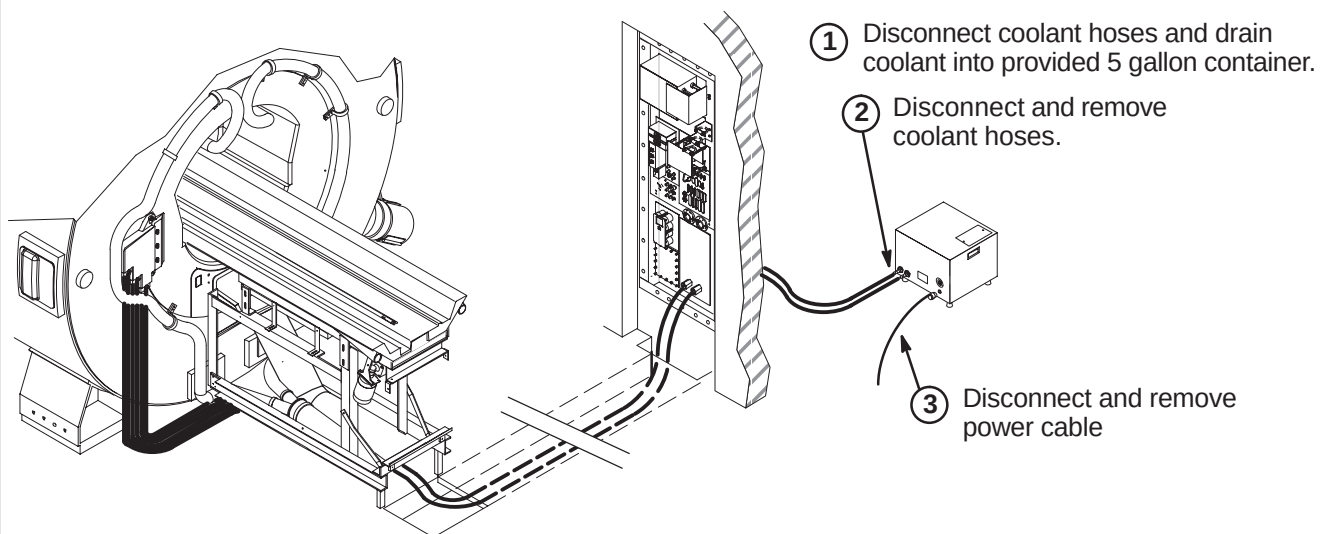
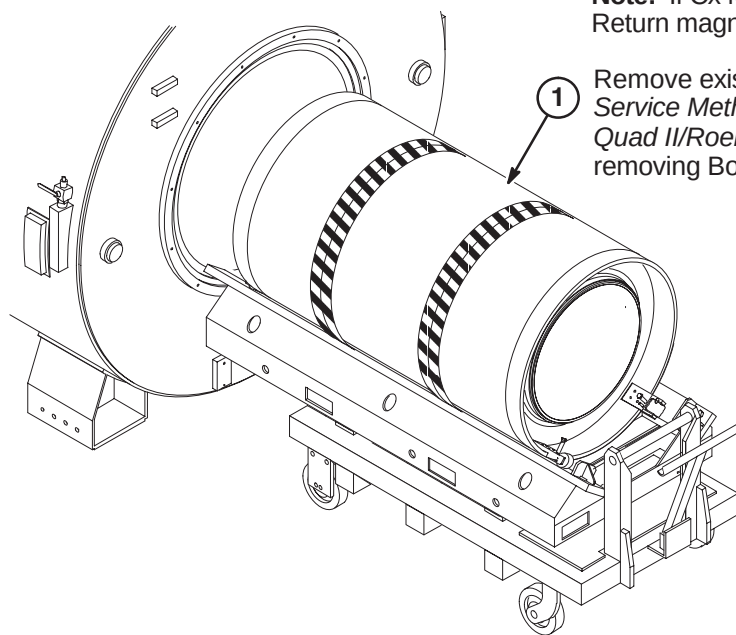
5-5 RETURN OF DEINSTALLED MATERIAL

ITEMS TO BE RETURNED FOR RECYCLING

Note: All items de-installed during this upgrade are to be returned for recycling. Secure as many of the items as possible to the two shipping dollies provided for this purpose. Shown below is an example of how the dollies might look when loaded. (not all items are shown)



5-6 DEINSTALL BODY COIL (EXCEPT FOR Cx MAGNETS)

☐ DRAIN AND REMOVE HEAT EXCHANGER (Horizon Systems)☐ REMOVE BODY COIL

Note: If Cx Magnet, do not remove Body Coil. Return magnet with Body Coil instact.

① Remove existing Body Coil. Refer to procedures in *MR Service Methods CD-ROM, Replacement: Gradient Coil, Quad II/Roemer Coil 1.5T Replacement* for instructions on removing Body Coil.

Note: If Body Coil is shipped back to GE Renewable Resources in the magnet bore, then 2 x 4 bracing must be installed across the bore opening and blocks put in place between the ends of the coil and the bore bracing to prevent the coil from moving during shipment. See instructions on page 5-5.

5-6 DEINSTALL BODY COIL (EXCEPT FOR Cx MAGNETS) (continued)

BRACING BODY COIL WITHIN MAGNET BORE (Alternative)

① To create the bracing for returning the Body Coil in the magnet you must obtain locally:

- A. 2 x 4 framing lumber to create Body Coil braces for front and rear of magnet.
- B. Four $\frac{3}{8}$ -16 x $2\frac{1}{2}$ long Hex Head Screws.
- C. At least twelve $\frac{3}{8}$ Washers for above hex screws.
- D. Eight $\frac{5}{16}$ or $\frac{3}{8}$ Lag Bolts for attaching short lengths to 2 x 4 spanning the bore.
- E. At least eight washers for the lag bolts.

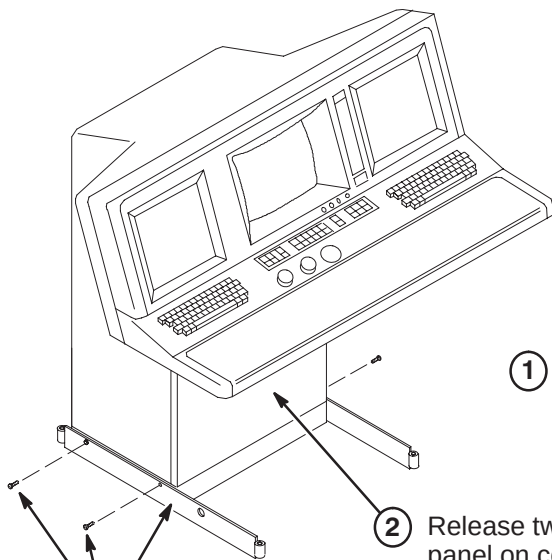
② Cut 2 x 4 to a length long enough to span mounting holes in magnet end ring. Drill two holes and attach to end ring using $\frac{3}{8}$ -16 x $2\frac{1}{2}$ long Hex Head Screws and washers.

③ Cut 2 x 4 braces to fit between end of Body Coil and end ring cross piece. Attach with two lag bolts and washers for each brace as shown.

④ Repeat process for other end of magnet.

5-7 DEINSTALL 3.x/4.x OPERATOR CONSOLE RETURN PROCEDURE

Confirm that all cables are disconnected from Operator Console. Perform the following procedure for return of 3.x/4.x Operator Console.

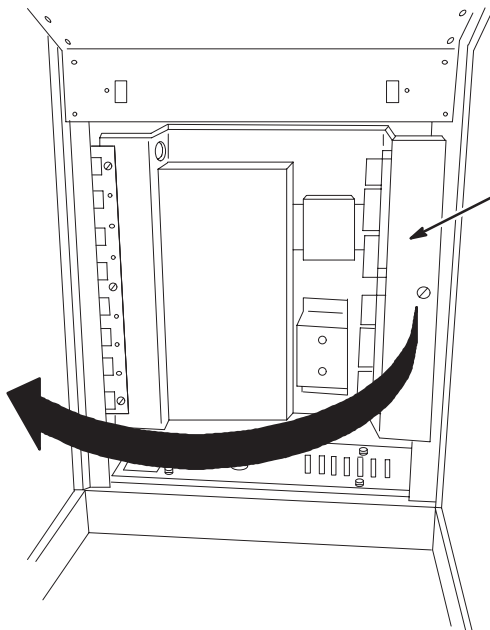
□ A1 – INSTALL OPERATOR CONSOLE SHIPPING LEGS

① Remove any installed Service Keys.

② Release two captive fasteners near top of lower front panel on console and remove panel.

③ Fasten to each side of Console:

- (1) 46-225881G1 Anti-Tip Shipping Leg
- (2) 46-208561P19 1/4-20x 3/4 Long Hex Head Screw

□ A2 – OPERATOR CONSOLE REMOVAL

① Unlatch and swing open panel assembly.

② Remove hardware that attaches console to floor.

③ Close and latch panel assembly.

④ Re-attach Front Panel.

5-8 DEINSTALL 5.x OPERATOR CONSOLE AND RETURN PROCEDURE

Confirm that all cables are disconnected from Operator Console. Perform the following procedure for return of 5.x Operator Console.

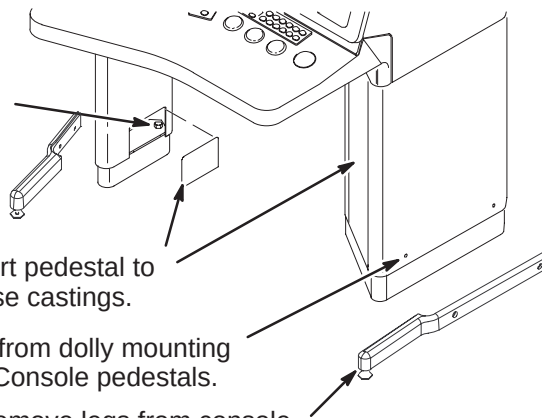
□ A1 – PREPARING OPERATOR CONSOLE REMOVAL

- ④ Loosen screws holding Console to pedestal risers.

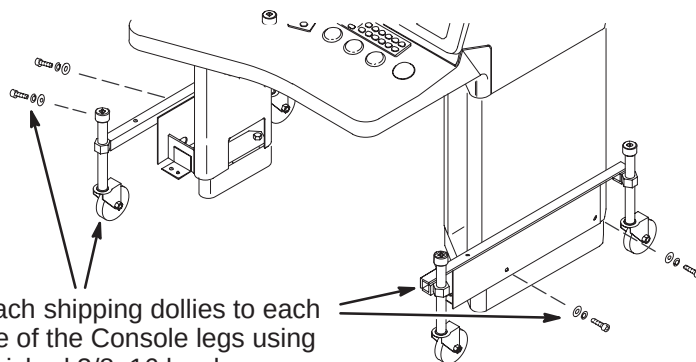
- ③ Remove covers on sides of support pedestal to access mounting holes on the Base castings.

- ② Remove Plug Buttons from dolly mounting holes on each side of Console pedestals.

- ① If installed, unfasten and remove legs from console.

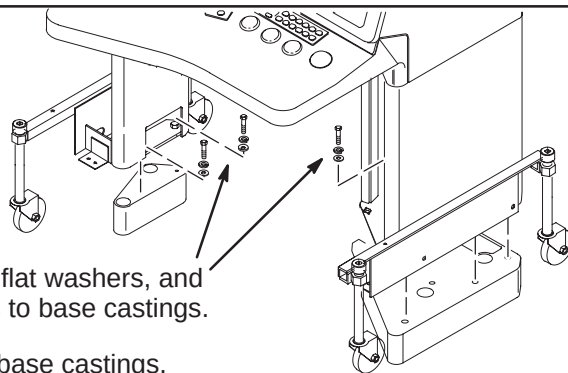
**□ A2 – ATTACHING DOLLY TO OPERATOR CONSOLE**

- ① Attach shipping dollies to each side of the Console legs using furnished 3/8–16 hardware.

**□ A3 – REMOVING OPERATOR CONSOLE**

- ① Remove lock washers, flat washers, and screws connecting legs to base castings.

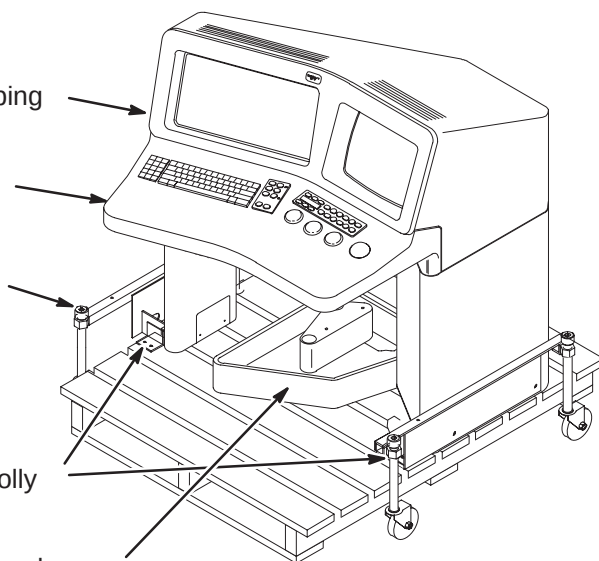
- ② Raise console up over base castings.



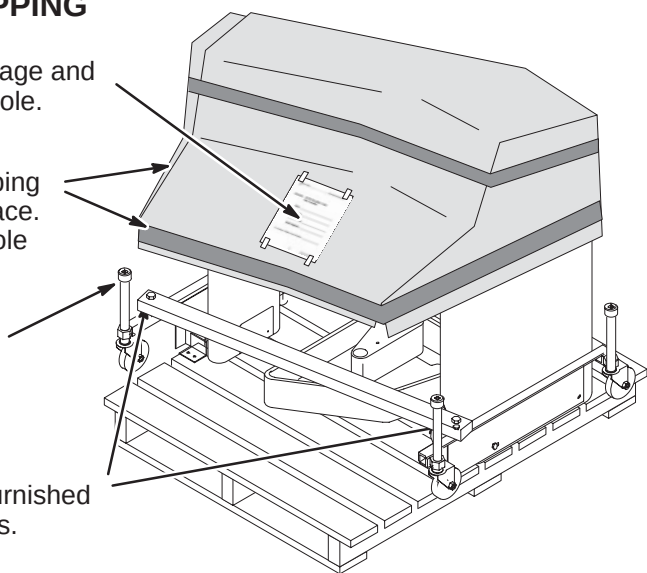
5-8 DEINSTALL 5.x OPERATOR CONSOLE AND RETURN PROCEDURE (Cont'd)

□ A4 – FASTENING OPERATOR CONSOLE TO SHIPPING PALLET

- ① Carefully roll Operator Console over an available shipping pallet.
- ② Position Console so that Keyboard Housing does not extend beyond shipping pallet.
- ③ Use a 1/2 inch socket wrench to turn the jack screws and lower the Operator Console onto pallet.
- ④ Locate (4) four provided 5/16 x 1-1/2 long lag bolts.
- ⑤ Install two of the lag bolts thru each shipping dolly bracket to fasten to shipping pallet.
- ⑥ Nest together Riser castings on pallet as shown.

**□ A5 – FINAL PREPERATION BEFORE SHIPPING**

- ④ Fill out *GEMS Recycling Form* on next page and attach to top of cardboard covering console.
- ③ Using any available cardboard (i.e. empty shipping boxes), cover Operator Console and tape in place. This is to prevent damage when shipping console back to manufacturer.
- ② Make sure that all four wheel assemblies are completely raised. If not, use 1/2 inch socket wrench to turn jack screws.
- ① Install provided dolly stiffener with furnished 1/4-20 x 1-1/2 long screws and nuts.



Return Console to GE Medical Systems with other system equipment that will be replaced during this upgrade.

Attn: GEMS GoldSeal

FDO: _____

Site Name: _____

Removal Date: _____

Field Engineer: _____

Genesis Console Serial Number: _____

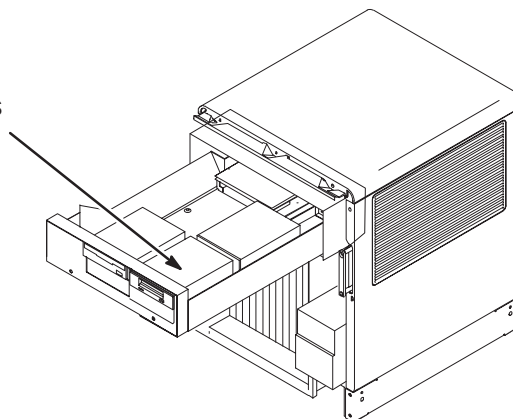
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5-9 DEINSTALL 5.x COMPUTER AND RETURN PROCEDURE

A1 – REMOVAL OF DAT AND MOD DRIVES FROM GENESIS COMPUTER

UPGRADE FROM 5.x OPTIONS:

Remove Legacy Media DAT and MOD Drives from Genesis Computer for transfer to new Operator Workspace computer.



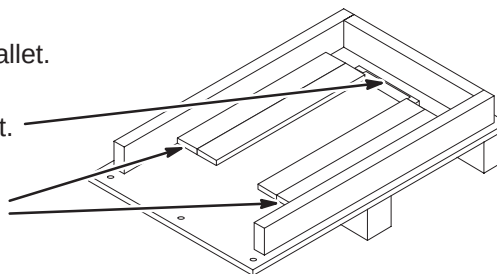
Confirm that all cables are disconnected from Computer. Perform the following procedure for return of 5.x Computer reusing the shipping pallet that the new Operator Workspace Cabinet arrived in.

A2 – PREPARING SHIPPING PALLET

① Locate the Operator Workspace Cabinet shipping pallet.

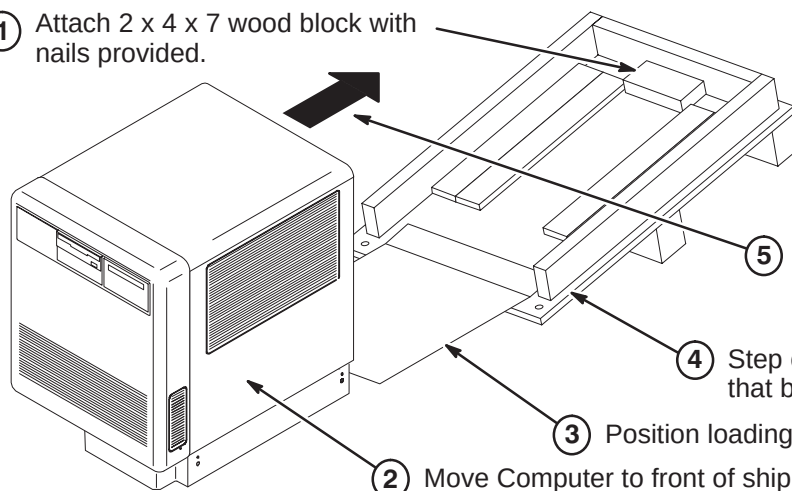
② Remove existing bracket.

③ Re-position the protective foam on each side from a vertical to a horizontal position between the side of the pallet and the existing horizontal foam.



A3 – LOADING COMPUTER ONTO SHIPPING PALLET

① Attach 2 x 4 x 7 wood block with nails provided.



Note:

Pallet will return to horizontal position as Computer is rolled toward back of pallet.

⑤ With another person, carefully roll Computer onto shipping pallet so it will rest on the horizontal foam pieces.

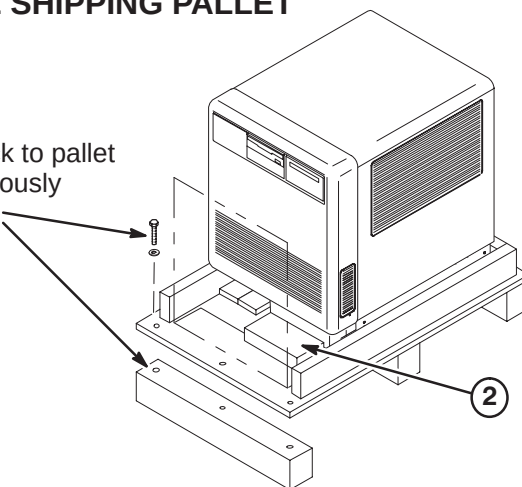
④ Step on front corner of shipping pallet so that bottom board of pallet touches floor.

③ Position loading board at front of pallet.

② Move Computer to front of shipping pallet.

5-9 DEINSTALL 5.x COMPUTER AND RETURN PROCEDURE (Cont'd)

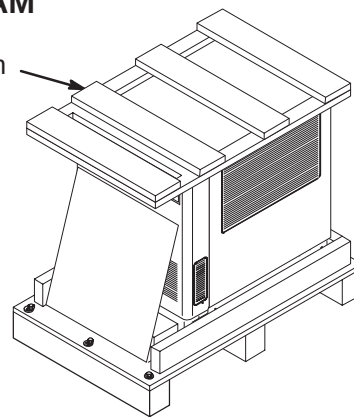
□ A4 – REASSEMBLE SHIPPING PALLET

- ① Re-attach wood block to pallet using the three previously removed fasteners.
- ② Place 2 x 4 wood block against front right side of Computer and attach to pallet with nails provided.
- 

□ A5 – INSTALLING COMPUTER TOP SHIPPING FOAM

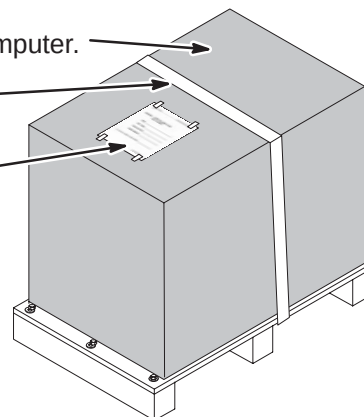
- ① Install the Operator Workspace Cabinet shipping foam on top of Computer.

Note: Shipping foam insert will not fit as well as it did with the new Operator Workspace Cabinet.



□ A6 – FINAL PREPERATION BEFORE SHIPPING

- ① Place Operator Workspace shipping box over Computer.
- ② Secure to shipping pallet.
- ③ Fill out *GEMS Recycling Form* on next page and attach to top of shipping box.



Return Computer to GE Medical Systems with other system equipment that has been replaced during this upgrade.

Attn: GEMS GoldSeal

FDO: _____

Site Name: _____

Removal Date: _____

Field Engineer: _____

Genesis Computer Serial Number: _____

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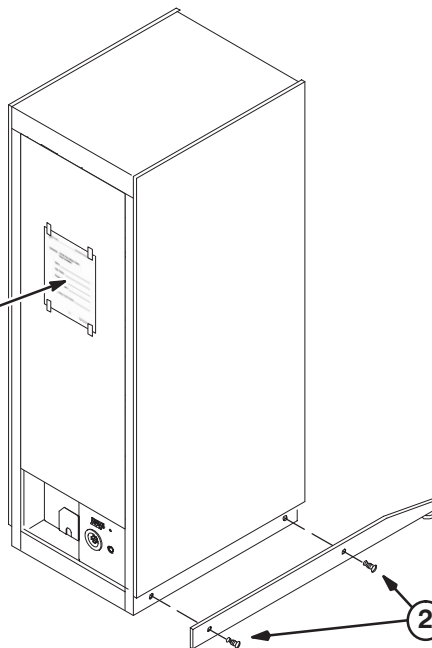
5-10 CABINET RETURN PROCEDURE

Confirm that all cables are disconnected from applicable Cabinet. The System Cabinet is shown below. The following procedure should be used for return of any cabinet with anti-tip bars.

Refer to following deinstallation for cabinets being removed

REMOVING ANTI-TIP BARS

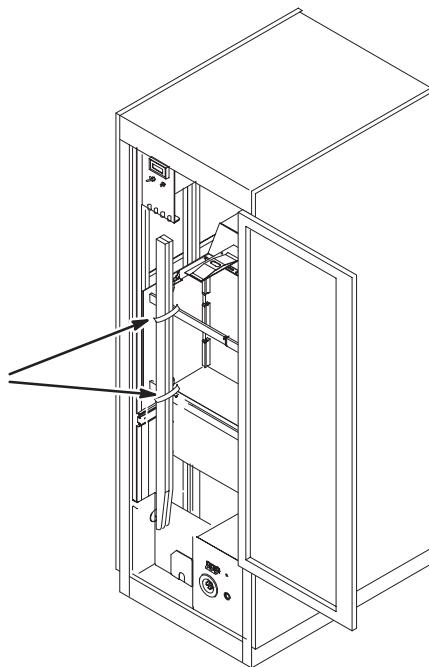
- ① Fill out *GEMS Recycling Form* on next page and attach to outside of cabinet door.



- ② Remove anti-tip bars from cabinet.

ATTACHING ANTI-TIP BARS AND FORM FOR RETURN SHIPMENT

- ① Fasten anti-tip bars to inside of cabinet for shipment back to GE Medical Systems.



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NOTE: Copy this sheet as needed.

Attn: GEMS GoldSeal

FDO: _____

Site Name: _____

Removal Date: _____

Field Engineer: _____

Cabinet Type: _____

Cabinet Serial Number: _____

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5-11 DEINSTALL SYSTEM CABINET**□ SYSTEM CABINET UPGRADE GUIDELINES**

The Signa Horizon 8.x System Cabinet Upgrade includes a new cabinet which will require items to be transferred from the old cabinet. This Upgrade also provides instructions for other components and options being transferred to the new System Cabinet. Note that various parts will be left over according to the type of system being upgraded. Return to GE these parts.

UPGRADE FROM SIGNA 3.x SYSTEM CABINET

Any Signa 3.x System being upgraded to Signa Horizon 8.x HiSpeed, or EchoSpeed will receive a new System Cabinet with all components installed.

UPGRADE FROM SIGNA ADVANTAGE 4.x SYSTEM CABINET

Any Signa Advantage 4.x System being upgraded to Signa Horizon 8.x HiSpeed or EchoSpeed will receive a new System Cabinet without the TNF Module.

- The new Horizon 8.x System Cabinet does not have a TNF Assembly installed. If TNF Module assembly is present in the old two-bay System Cabinet, it will need to be transferred. Refer to procedure on next page for removal. If not present Catalogs M1090ZB or M1090ZF should have been ordered prior to upgrade.

UPGRADE FROM SIGNA ADVANTAGE 5.x SYSTEM CABINET

Any Signa Advantage 5.x System being upgraded to Signa Horizon 8.x HiSpeed or EchoSpeed will receive a new System Cabinet without the TNF Module.

- The new Horizon 8.x System Cabinet does not have a TNF Assembly installed. If TNF Module assembly is present in the old Advantage 5.x System Cabinet, it will need to be transferred. Refer to procedure on next page for removal. If not present Catalogs M1090ZB or M1090ZF should have been ordered prior to upgrade..

UPGRADE FROM SIGNA HORIZON 5.x SYSTEM CABINET

Any Signa Horizon 5.x System being upgraded to Signa Horizon 8.x HiSpeed or EchoSpeed will receive a new System Cabinet without the TNF Module.

- The new Horizon 8.x System Cabinet does not have a TNF Assembly installed. If TNF Module assembly is present in the old Horizon 5.x System Cabinet, it will need to be transferred. Refer to procedure on next page for removal. If not present Catalog M1090ZB, M1090ZC, M1090ZF, or M1090ZG should have been ordered prior to upgrade.
- For upgrading from a Horizon 5.x EchoSpeed to a Horizon 8.x Echospeed, the Fast Receiver and cable (2137306) will need to be transferred, refer to procedures on next page for removal.

UPGRADE FROM SIGNA HORIZON 8.x SYSTEM CABINET

Any Signa Horizon 8.x System being upgraded to Signa WideOpen 8.x HiSpeed or EchoSpeed will have Magnet Monitoring installed.

- Remove cryogen meters and related cables. Return with magnet.

5-11 SYSTEM CABINET REMOVAL (continued)

REMOVE TNF MODULE FROM EXISTING CABINET

For Upgrading from Horizon 5.x: TNF Assembly is present in existing cabinet for transfer. The TNF Expansion Asm (2118107) should also have been installed if Multi-coil is being transferred.

For Upgrading from Advantage 5.x: If present, transfer TNF Assembly Module from existing cabinet. If TNF Assembly Module is not present, order 2118106 for non Multi-coil system or 2118106 and 2118107 for Multi-coil.

For Upgrading from Advantage 4.x: If present, transfer TNF Assembly Module from existing cabinet. If TNF Assembly Module is not present, order 2118106 for non Multi-coil system or 2118106 and 2118107 for Multi-coil.

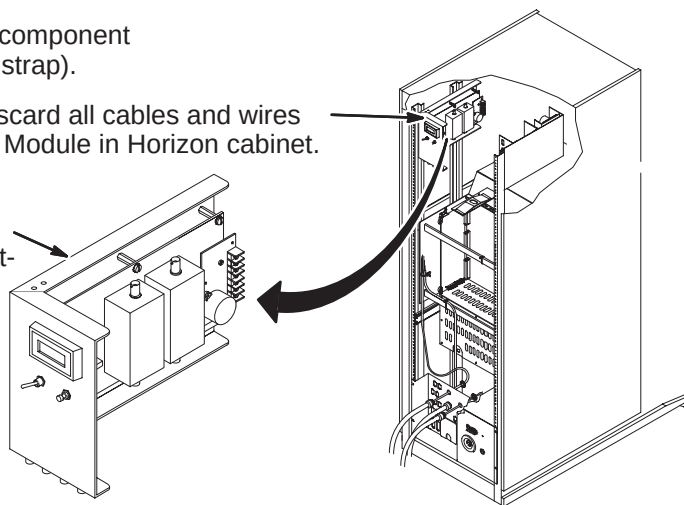
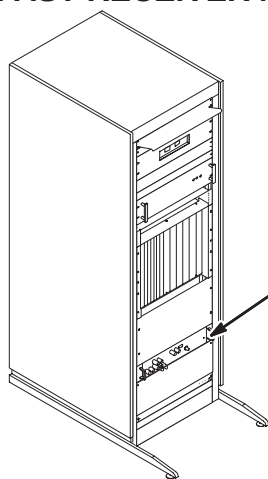
For Upgrading from Advantage 3.x: TNF Assembly should be present in new 8.x cabinet. The TNF Expansion Asm should also have been installed if Multi-coil is part of new 8.x system. If not present, order 2118107.



Note: TNF Module is a static sensitive component and needs to be handled as such (wrist strap).

- 1 Disconnect and discard all cables and wires connected to TNF Module in Horizon cabinet.

- 2 Unfasten and remove TNF Assembly Module (with Expansion Asm if present). Save mounting hardware.

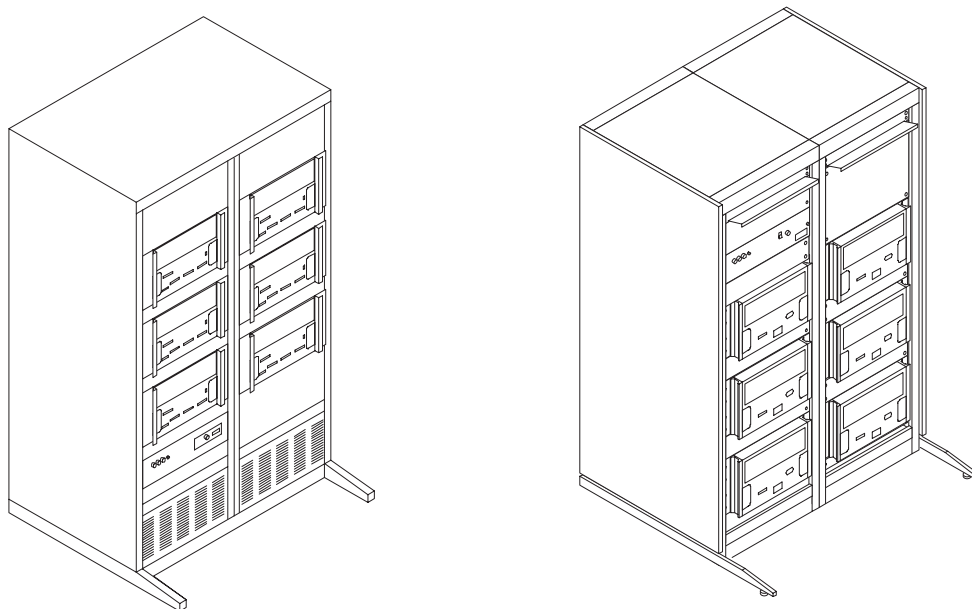
**REMOVE FAST RECEIVER MODULE FROM HORIZON SYSTEM CABINET**

- 1 Remove Fast Receiver, rear and front vertical mounting rail brackets, mounting hardware, and slides from Horizon 5.x System Cabinet.

5-12 DEINSTALL PRE-8645 GRADIENT CABINET

☐ PRE-8645 GRADIENT CABINET REMOVAL

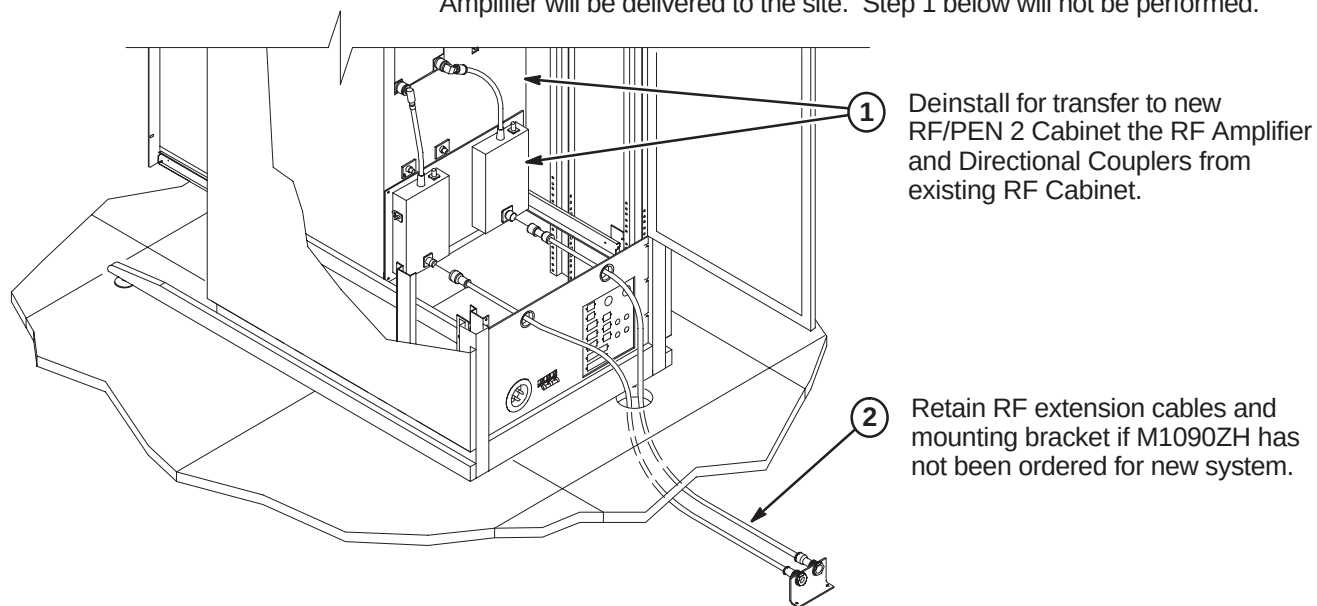
- ① Disconnect all cables. Cabinet to be returned.



5-13 DEINSTALL RF CABINET

☐ D – RF CABINET CABLE CONNECTIONS

Note: For a 3.x to 8.x WideOpen upgrade, a new RF/Pen Cabinet with RF Amplifier will be delivered to the site. Step 1 below will not be performed.



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5-14 DEINSTALLATION FEEDBACK DATA SHEET

Use this form to list the problems you had with the Deinstallation at this site.

Site Name: _____

Location: _____

Magnet Type: _____ (S2, S3, S4 etc...)

System Type: _____ (3x, 4x, 5x, other)

Problems encountered with desinstall procedures and Flowcharts

Site related problems

List any cables that were cut or damaged during deinstall. List any hardware that sustained any major damage during deinstall:

Send this completed form to:

Magnet Deinstall Feedback
MR Service Engineering, W808
GE Medical Systems
P.O. Box 414
Milwaukee, Wisconsin 53201-9824